A 112-Gb/s PAM-4 Long-Reach Wireline Transceiver Using a 36-Way Time-Interleaved SAR ADC and Inverter-Based RX Analog Front-End in 7-nm FinFET

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***Abstract*— A 36-way time-interleaved 56-GS/s 7-bit ADC is designed to realize 112-Gb/s pulse-amplitude modulation (PAM-4) transceiver in a 7-nm FinFET CMOS. The receiver analog front-end stages and the ADC track-and-hold (T/H) buffers are implemented using inverter-based Gm/inverse-Gm-load cells. A distributed inductor peaking network and multi-phase clock calibration is implemented in the quarter-rate transmitter. The transceiver achieves <1E-8 pseudorandom binary sequence (PRBS)-31 PAM-4 bit error rate (BER) over a channel with 37.5-dB loss at 28 GHz while dissipating 602 mW per channel, excluding DSP.**

***Index Terms*— 112 Gb/s, 7-nm FinFET, CMOS, DSP, inverter-based amplifier, pulse-amplitude modulation (PAM-4), time-interleaved SAR ADC, wireline transceiver.**

I. Introduction

INCREASING demand for more bandwidth from data center and communication applications has been accelerating the development of 112-Gb/s transceivers in recent years [1]–[5]. Pulse-amplitude modulation (PAM-4) signaling remains the dominant choice of modulation scheme due to its superior spectral efficiency when compared with nonreturn-to-zero (NRZ) modulation and its simpler implementation when compared with more sophisticated schemes. In many applications, a multitude of these transceivers are monolithically integrated in ASICs and FPGAs to realize high aggregate throughput per chip, therefore mandating the power and area efficiency to be the primary design optimization goal. In addition, these high-throughput VLSIs tend to adopt more advanced process design kits (PDKs) to benefit from the technology scaling that enables compact and energy-efficient implementations of their processing logic cores. While the logic cores benefit from advanced process nodes optimized for digital designs, high-speed transceiver design flows need to commence with a consideration of the constraints and capabilities of a given PDK to result in an efficient solution.

This article reports a long-reach capable 112-Gb/s PAM-4 transceiver designed in a 7-nm FinFET process, extensively using CMOS-style circuit implementations of high bandwidth linear and nonlinear functions needed for data paths and clock paths. Specifically, the entire receiver analog front-end (AFE) and ADC sampling front-end (SFE) are realized by inverter-based gain stages and complementary CMOS pass-gates, obviating the need for current sources and boot-strapped switches. High-speed clock distribution, multi-phase clock generation, duty cycle distortion (DCD)/IQ correction, and phase rotation circuits are also implemented exclusively in the CMOS style. The sub-ADC is implemented in the digital-process-friendly SAR ADC topology. The decision of sub-ADC circuit specification and time-interleaving architecture of the ADC SFE is optimized to fully use the device f_T of the 7-nm FinFET process, to achieve an overall area and power efficiency customized for ultra-high-speed wireline receiver applications.

II. Transceiver Architecture

Fig. 1 illustrates the top-level block diagram of the transceiver, consisted of a shared LC-phase-locked loop (PLL), a transceiver channel, and a control block. Each channel includes a TX, an RX, and a shared channel clock generation block. A shared bias generation and a resistor calibration block are also included. The control block generates a pseudorandom binary sequence (PRBS) pattern, sends 875-MHz 128-b-wide TX data, receives 778-MHz 144-b-wide RX data to and from transceiver channels, and calculates the bit error rate (BER). It includes a raw ADC data capture buffer and interfaces with an off-chip FPGA controller running a background RX adaptation and various calibration loops.

The LC-PLL synthesizes a 14-GHz differential clock from a 583.33-MHz external reference clock. The PLL output clock is

Manuscript received May 1, 2020; revised July 9, 2020; accepted September 2, 2020. Date of publication September 25, 2020; date of current version December 24, 2020. This article was approved by Guest Editor Friedel Gerfers. *(Corresponding author: Jay Im.)*

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Digital Object Identifier 10.1109/JSSC.2020.3024261

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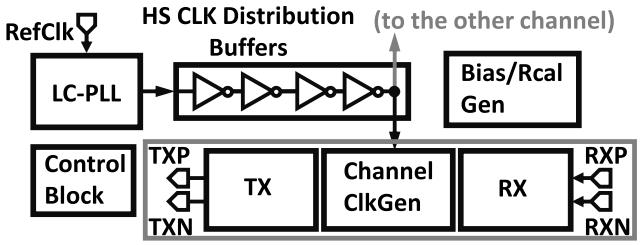


Fig. 1. Transceiver top-level block diagram. (One transceiver lane is shown).

then delivered to each of the two channels via supply-regulated CMOS high-speed clock distribution buffers. The third-order analog LC-PLL uses a compact wide-range transformer-based LC-voltage-controlled oscillator (VCO) [9], optimized for a lower jitter needed for 112-Gb/s operation.

The transceiver design is completed with production specification verified and signed off, including process, voltage, and temperature (PVT) tolerance, timing margin, and reliability checks (electro-migration, device self-heating, bias temperature instability (BTI), HCI, time-dependent dielectric breakdown (TDDB), and electrostatic discharge (ESD), latch-up). Notably, high-bandwidth RX AFE and ADC SFE circuits use thin-oxide MOSFETs for best gain-bandwidth product and f_T available from the 7-nm FinFET PDK. Their device reliability is guaranteed by carefully designing dc bias points both in the normal operation mode and in power down mode such that device terminal voltage differences do not exceed the allowed maximum rating for thin-oxide devices.

III. Transmitter

The TX architecture shown in Fig. 2 adopts a four-tap feed-forward equalizer (FFE) (for two pre-cursor and one post-cursor taps) and a quarter-rate clocking scheme with duty cycle and I/Q mismatch calibration to enable the use of a power-efficient CMOS 4:1 MUX [4]. A current-mode logic (CML) pad driver topology [4], [5] is chosen to achieve 3% finite-impulse response (FIR) resolution without incurring an excessive power penalty from the large number of unit cells containing high-bandwidth node data switches as would be required in a voltage-mode (VM) driver. An array of tail-less CML driver cells with source node switching topology allows the use of more power-efficient CMOS pre-driver and 4:1 MUX. The data path comprises a 128 to 32 serializer, a four-tap FFE data generation, and a 32 to 8 serializer, followed by the final 4:1 MUX. A retimer array is needed to close the timing between the 32:8 serializer and the 4:1 MUX array due to the relatively long routings between the two blocks.

A. Final 4-to-1 MUX

The final 4:1 MUX stage is a performance-critical block in quarter-rate TX designs as it has the tightest timing paths, hence it largely determines the transmitter output jitter performance. Consequently, the block contributes a significant amount to the total TX power consumption. We implemented the 4:1 MUX using a simple CMOS pass-gate-based two-stage MUX clocked by four-phase 14-GHz clocks as shown in Fig. 3. The use of CMOS MUX is motivated by its compact layout and superior energy efficiency.

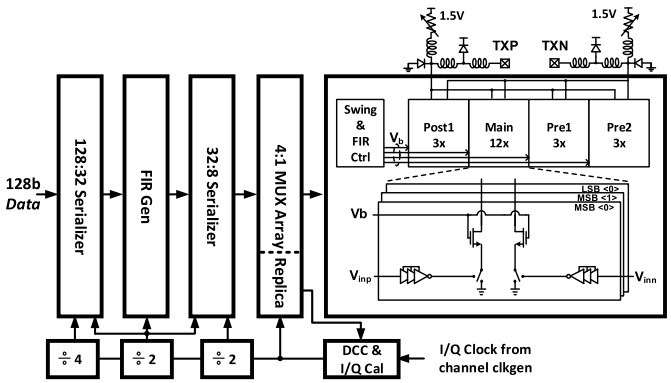


Fig. 2. TX sub-system.

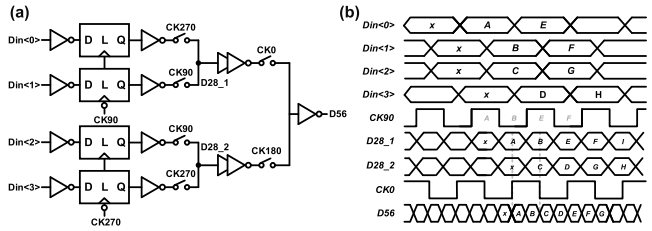


Fig. 3. (a) TX final 4:1 MUX. (b) Timing diagram.

The 14-GHz I/Q clock is produced by the channel clock generation block shared by TX and RX and distributed to TX in full CMOS swing via a regulated CMOS buffer chain. A TX clock calibration block for duty cycle correction (DCC) and I/Q error compensation is needed to correct the duty cycle distortion and I/Q phase errors caused by device mismatches from the MUX circuit so that a low deterministic jitter can be achieved at 112 Gb/s. The phase errors are sensed at the output of the replica MUXs and are background calibrated with smaller than 100-fs resolutions.

B. Pad Driver

A tail-less CML driver [5] is chosen in this design to achieve four-tap FIR (two pre-cursors and one post-cursor) with less than 3% step size. The CML driver is known to consume 4 times more current than the VM driver for the same output swing. However, a CML driver can attain a fine FIR resolution more easily by controlling its DC biasing without affecting high-speed signal paths. In contrast, a slice-based VM driver [4] implements the FIR by configuring the input data of each unit slice, which limits the minimum number of required 4:1 MUXes. Consequently, the amount of power saved in the VM pad driver is not compensated enough for the power overhead of the 4 to 1 MUX and clocking circuits in a quarter-rate architecture with a fine FIR resolution.

In this work, a distributed peaking network is designed to compensate for over 200-fF device and parasitic capacitance at the driver current summing node. A T-coil with an additional series inductor effectively separates the N- and P-type ESD diodes, C4 bump pad, and on-die termination (ODT) resistor array, implementing a lumped transmission line for bandwidth extension and improved return loss. With the distributed peaking network, a 46-GHz bandwidth is achieved at the TX output.

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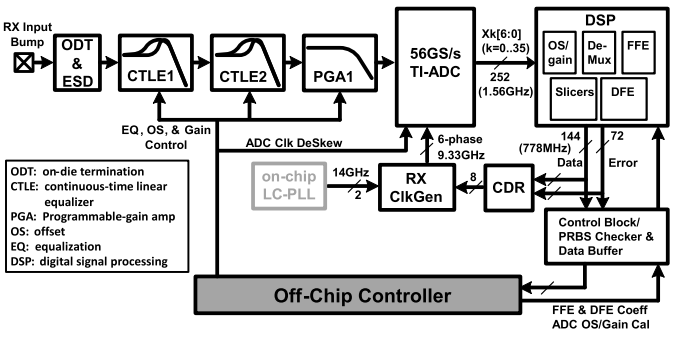


Fig. 4. RX sub-system.

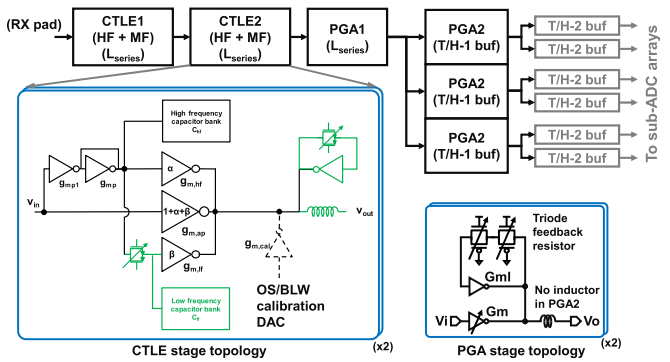


Fig. 5. RX analog front-end implemented using inverter-based Gm/Gml stages.

IV. Receiver

Fig. 4 depicts the RX top-level functional diagram. The PAM-4 signal received at the RXP/RXN input pads is terminated by calibrated ODT resistors. T-coil inductors are used for impedance matching to improve return loss and bandwidth by neutralizing parasitic capacitances caused by input pads, ESD diodes, ODT, and continuous-time linear equalizer (CTLE) input capacitance. The all CMOS RX AFE, consisting of two stages of CTLE, equalizes the input signals by boosting the Nyquist frequency (HF) and mid-range frequency (MF) bands separately. The following two stages of the programmable-gain amplifier (PGA) adjust aggregated broadband gain for automatic gain control adaptation that controls the output signal range for the subsequent ADC to digitize with fully used full scale range (FSR).

The ADC SFE realizes analog signal de-muxing by two track-and-hold (T/H) stages. The second PGA stage acts as the first-rank T/H buffer. A two-rank 6-by-6 time interleaving serially breaks down the ADC input signal into 36 sub-ADCs, which subsequently digitize the signal into 7-b symbols, xk[6:0] ($k = 0..35$). The 6×6 interleaver architecture is chosen with considerations of tradeoffs among the SFE, clocking circuitry, and sub-ADCs' sampling speeds, which will be discussed in further detail.

The DSP retimes ADC output symbols, xk[6:0] ($k = 0..35$), and performs offset and gain correction for the 36 individual sub-ADCs. The 72 deserialized xk are fed into subsequent 31-tap FFE and one-tap speculative decision-feedback equalizer (DFE) stages for ISI cancellation before being converted to produce data and error bits.

An on-chip baud-rate clock and data recovery (CDR) logic detects the optimal ADC sampling aperture phase (T/H-1 clock falling edge) by accumulating the Mueller–Muller [8] phase detector output to update an 8-b phase interpolator (PI) code.

A. Data Path

The RX AFE in Fig. 5 consists of a two-stage CTLE followed by two PGA stages. The CTLEs implement high-frequency peaking with a feedforward pole-subtractive topology, and hybrid peaking (i.e., both HF and MF) is realized in both the stages [8]. At the max peaking setting, the CTLE provides more than 16 dB of peaking at the Nyquist frequency. MF tuning provides additional long-tail ISI cancellation, which helps reduce the post-cursor FFE coefficient ranges. To further reduce the area of the device capacitors for MF path, a series triode device is inserted to allow resistance tuning for the zero frequency. A combination of active-inductor Gm load (Gml) and passive inductor effectively boosts the bandwidth of each stage of CTLE and PGA1. Compact inverter-based amplifier stages consisted of a Gm cell and inverse-Gml, with inherently good linearity due to ratio-metric design [8], are used to implement CTLE, PGA, and ADC T/H buffers. The active inductor is realized by complementary pass-gates in the triode mode, which are used as feedback resistors instead of the conventional TiN resistors, dramatically reducing the cell size and parasitic wire loading. All AFE circuit elements (Gm, Gml, triode-mode feedback resistors, zero resistors, and zero capacitors) are laid out using the same minimum design rule poly-to-poly pitch to achieve high area efficiency and more process uniformity. An overall 70% area reduction (passive inductor area not included) is achieved compared with [8] for each CTLE stage.

The time-interleaved ADC architecture is carefully chosen to optimize the overall power and area efficiency under various constraints of the design. Due to the CMOS nature of the AFE and ADC, the fan-out of each can be decided in a "digital" fashion. Once the design of the sub-ADC is optimized, its input capacitance propagates eventually back to the RX input, determining along the way the drive strengths of the preceding stage given their respective bandwidth requirements. Therefore, we first evaluate the number of required sub-ADCs to minimize fan-out and reduce the overhead incurred by layout complexity and parasitics. To fully take advantage of technology scaling provided by the 7-nm FinFET, the sub-ADC is optimized for sampling speed and area.

The metastability impact on the sub-ADC performance could become the gating factor of the overall system performance. Therefore, an in-depth study was done to understand the effect of metastability-induced errors in asynchronous SAR ADCs [10]. While pushing the sub-ADCs to their speed limit can significantly increase the probabilities of metastability events, the generated errors still could be tolerable in SerDes applications. An intuitive understanding of the detailed analysis in [10] could be established with the following metastable conditions. In a high-speed asynchronous SAR ADC, it is more probable for the LSB bit position to run out of time for evaluation creating an LSB metastable event, but the error created is small (i.e., 0 or 1 LSB code error) and

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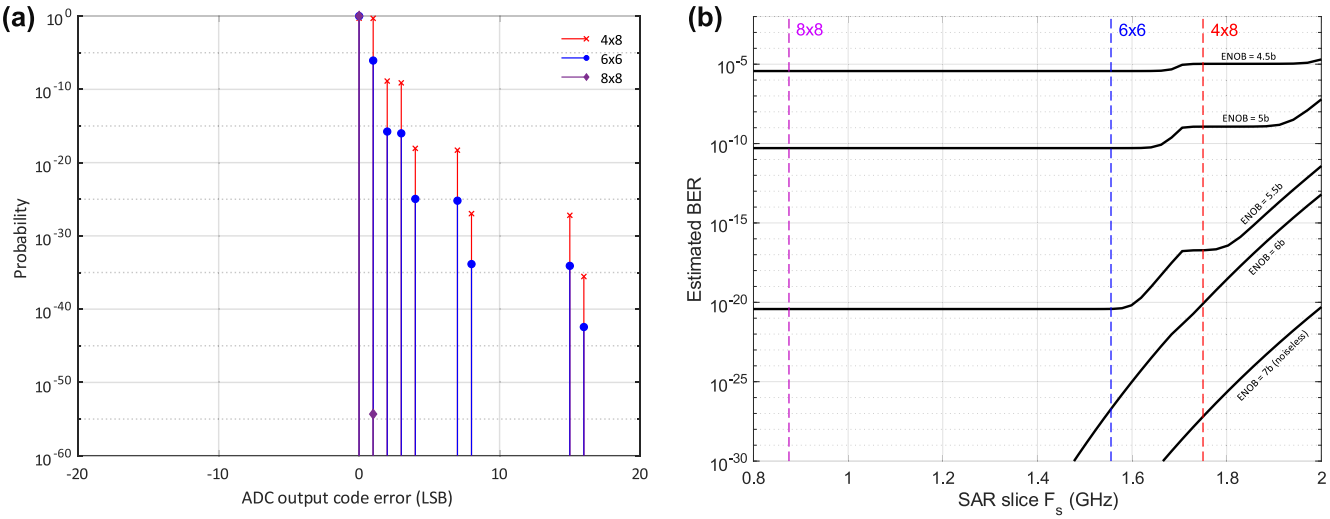


Fig. 6. (a) Simulated 7-b SAR ADC metastability error probability versus error magnitude for 4 × 8, 6 × 6, and 8 × 8 time-interleaving. (b) Simulated system BER versus sampling speed for five cases of thermal noise level.

typically below the thermal noise floor, thus having minimal impact on the BER of the overall system. On the other hand, an MSB bit position metastability event could yield much larger code errors, which almost certainly causes bit/symbol errors. However, these are exponentially unlikely events by construction of asynchronous SAR ADCs. Taking the 7-nm process capabilities and metastability impact on the system BER into account, we concluded that the sub-ADC design in this work can operate at around 1.6 GS/s across different PVT corners with a sufficient margin.

As an illustration, three possible time-interleaving architectures for 56 GS/s are considered for BER impact simulation: 4 × 8, 6 × 6, and 8 × 8, corresponding to sub-ADC sampling speed (F_S) of 1.75 GS/s, 1.56 GS/s, and F_S = 875 MS/s, respectively. Fig. 6(a) shows the simulated metastability error probability versus error magnitude in the unit of LSB for the three cases. The 4 × 8 case shows 1 LSB magnitude error probability almost at the same level as zero error, implying that the LSB bit resolution during the SAR loop almost always runs out of time. Also, 2 to 4 LSB error probabilities are high enough (∼1E-8) to impact on the system BER. On the other hand, the 8 × 8 case shows almost no metastability code errors with less than 1E-50 chance of failing with one LSB error. The 6 × 6 architecture achieves the balance between metastability error probabilities and pushing sub-ADC speed. The one LSB error probability is much smaller compared with that caused by the thermal noise, while higher impact 2 to 4 LSB errors are sufficiently rare (less than 1E-15 chance). Fig. 6(b) shows the simulated system BER versus F_S for five cases of system noise level as modeled by equivalent ENOB ranging from 4.5 to 7 b (noise-less). Three vertical dashed lines mark F_S corresponding to 8 × 8, 6 × 6, and 4 × 8 interleaving architectures, respectively. For 8 × 8, the system BER is always thermal noise limited (i.e., independent from F_S variation) and each SAR ADC slice is far away from their speed capabilities, under-utilizing 7-nm device speed limit. For the other extreme of 4 × 8 architecture, the metastability errors cause significant degradations in system performance, especially at lower baseline BER. For 6 × 6, we are operating closer to the metastability BER corners while the system performance is still dominated by thermal noise.

Given the suitable sampling speed of sub-ADC, various interleave architecture candidates are compared in Fig. 7. The 4 × 8 architecture stresses the sampling bandwidth of the front-end, clocking, and sub-ADC sampling speed despite its area advantage. An 8 × 8 architecture [2] significantly relaxes the speed requirement of each ADC component, but in turn wastes the extra speed offered by the process technology (e.g., concluded 1.6 GS/s capability of the sub-ADC with respect to the required 875-MS/s sampling speed). The area penalty and the associated floor-planning challenge of the 8 × 8 interleaving are also non-negligible due to a larger number of sub-ADC and clock generation. Therefore, the 6 × 6 architecture serves as a good trade-off between effective utilization of each key block design without over-stressing design margins and layout complexity. The performance-critical rank 1 T/H will sample at 9.333 GHz, which is a reasonable frequency for both the analog amplifiers and clock generation and distribution.

Fig. 8(b) illustrates detailed schematics of the 6 × 6 SFE. The SFE requires 3× rank 1 T/H buffers, each sampling with ∼50% non-overlapping clocks. The T/H-1 buffers also act as second-stage PGAs, which are a scaled down version of the inductor-less PGA1. Each T/H-1 buffer drives 2× rank 2 T/H unity gain buffers. All SFE stages use the same inverter-based circuits (Gm and active inductor triodes) as the CTLEs. The output common-mode of the Gm/Gml cells are compatible with the complementary CMOS pass-gates used for all T/H switches in the ADC, obviating the need for bootstrapping circuits and significantly reducing area and clock power consumption. A correct bias voltage is needed at the interface of rank 1 and rank 2 T/H buffers due to the nature of inverter-based T/H-2 Gm cells, and thus clock/charge injection switches are added for the T/H-1 switches. Since the StrongArm sense-amp comparator in

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	4x8	8x8	6x6 [this work]
Total # of sub-ADCs	32	64	36
Sub-ADC sampling speed	1.75GS/s	875MS/s	1.56GS/s
Rank1 sampler frequency	14GHz	7GHz	9.33GHz
Rank1 settling BW (7b)	~28GHz	~13GHz	~18GHz
# of T/H amplifiers	2 + 4	4 + 8	3 + 6
Area penalty	Small	Large	Medium
Clock distribution	Easy	Difficult	Moderate
Challenges	❑ Sub-ADC & rank1 speed ❑ Rank1 settling BW requirement	❑ Area & power penalty ❑ Clock distribution	❑ Manageable area & power penalty (compared to 4x8)
Advantages	❑ Compact layout ❑ Easy clocking	❑ Relaxed BW & timing requirement	❑ Good trade-off between process f_T & design margin

Fig. 7. Design tradeoffs for 4 × 8, 8 × 8, and 6 × 6 time-interleaving architectures for ADC SFE.

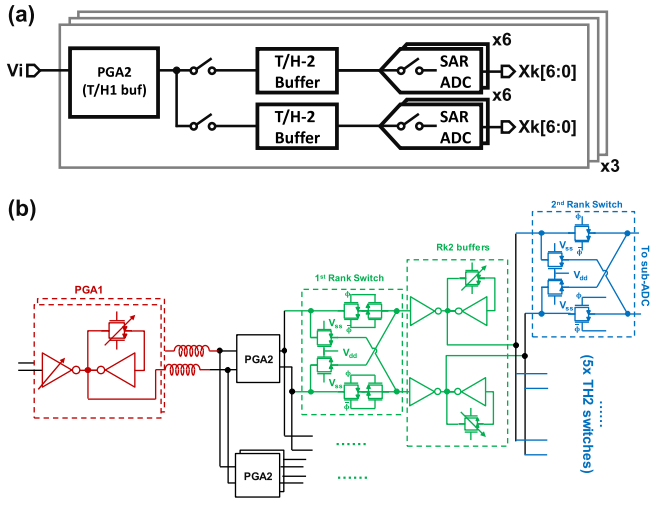


Fig. 8. (a) ADC sub-system and (b) circuit implementation of its sampling front-end.

sub-ADC can operate with a slightly wider common-mode range, the charge injection cancellation devices of T/H-2 switches are removed to reduce the parasitic capacitance at the capacitive digital-to-analog converter (CDAC) node. Both the stages have dummy cross-coupling devices for >50 dB hold mode feedthrough cancellation. The sub-ADC adopted a speed-optimized asynchronous SAR ADC architecture using a binary CDAC with an estimated unit capacitance of 0.5 fF.

B. Clock Generation

The RX clock generation architecture is driven from the top down by some key system constraints and optimizations. The use of a quarter rate architecture in the transmitter gives rise to the need for non-rotating 14-GHz quadrature input clocks, along with an LC PLL running at this frequency. The tradeoffs outlined regarding the speed of each sub-ADC and the number of sub-ADCs to optimize for area, power, and bandwidth result in the RX requirement for six equally spaced phases at 9.333 GHz, with optional phase rotation to track frequency differences between an external TX and the on-chip PLL. Generating the RX clocks from the same clock source as the TX obviates a need for a second clock source. The time-stamp information for a 9.333-GHz clock is already present in a 14-GHz clock, provided the duty cycle is accurate.

The architecture is given in Fig. 9. The 14-GHz clocks from the PLL are used as the injection for a four-stage CMOS 1:1 injection-locked oscillator (ILO). The four-stage ILO uses XOR-based phase sensing with supply regulation [7]. The outputs of this ILO drive both the TX and RX. At the RX, the eight equally spaced output phases at 14 GHz are used to drive an all CMOS 8-bit PI producing a rotating 14-GHz output. This differential clock has variable duty cycle control. The DCD-corrected output provides the source for an IQ-divide-by-6 circuit. The division by 6 is a cascade of a divide-by-3 and a divide-by-2, with one each in parallel for the I- and Q-paths. The quadrature is achieved by releasing the I-path first, followed by the Q-path 1.5 cycles later. The four-phase output at 2.333 GHz contains fewer transitions than a differential 14-GHz clock, thus being advantageous from a distribution power perspective. It is important to note that any DCD in the output of the PI will translate into the IQ error at the output of the divide-by-6 block. As such, IQ correction of the 2.333-GHz clocks is carried out by adjusting the duty cycle after the PI. An analog-style XOR detection is used to generate an error voltage which after processing with a dynamic comparator is used as an input to a digital control algorithm as illustrated in Fig. 9(c).

As illustrated in Fig. 10, the 2.333-GHz IQ clocks are now progressively processed to realize a multi-phase 9.333-GHz clock. The first step is a 4-to-1 self-timed multiplexer used to cyclically select the four 2.333-GHz clocks. This creates a 9.333-GHz pulse train with well-defined low jitter positive edges, but poorly controlled duty cycle. Passing this signal through a pulse generation circuit with adjustable pulse width, as shown in Fig. 10(a), generates a signal suitable for injecting into a ring oscillator pre-calibrated to near

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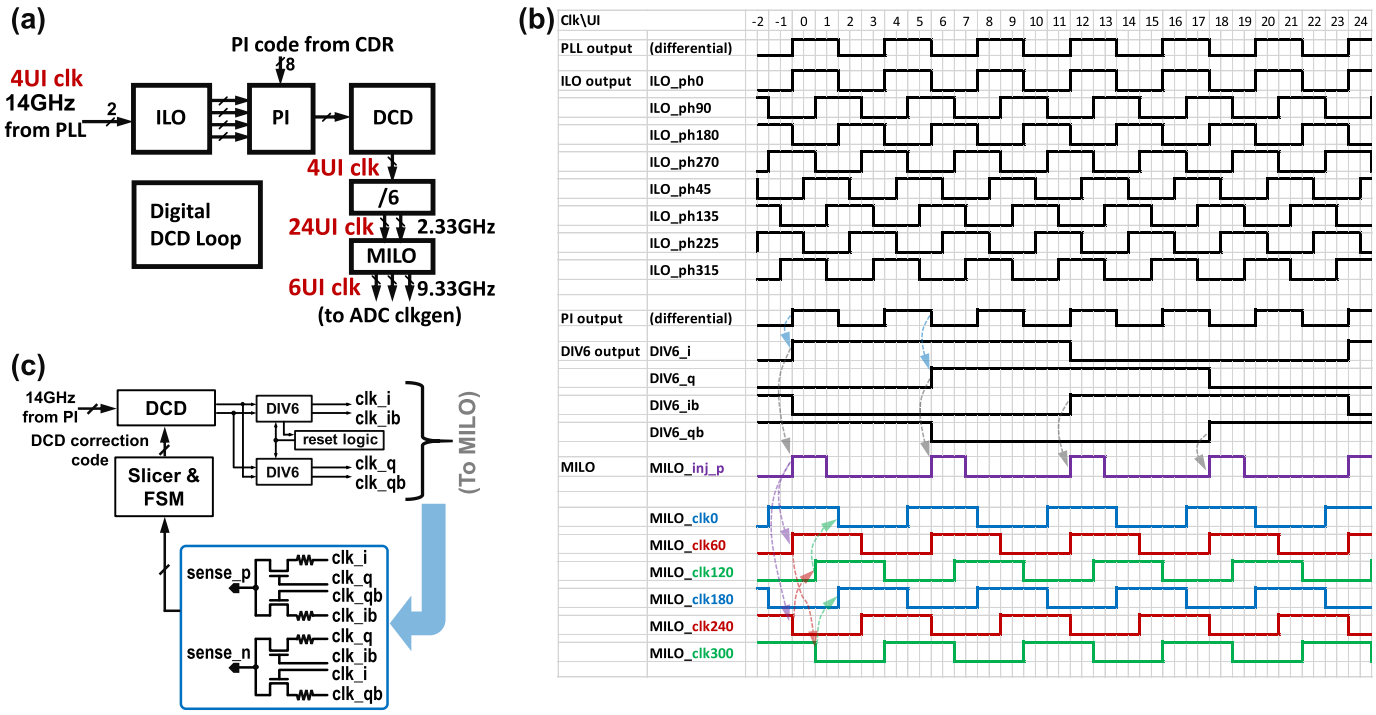


Fig. 9. (a) RX clock generation sub-system. (b) Timing diagram (only cycle-wise accurate). (c) DIV6 and DCD correction loop.

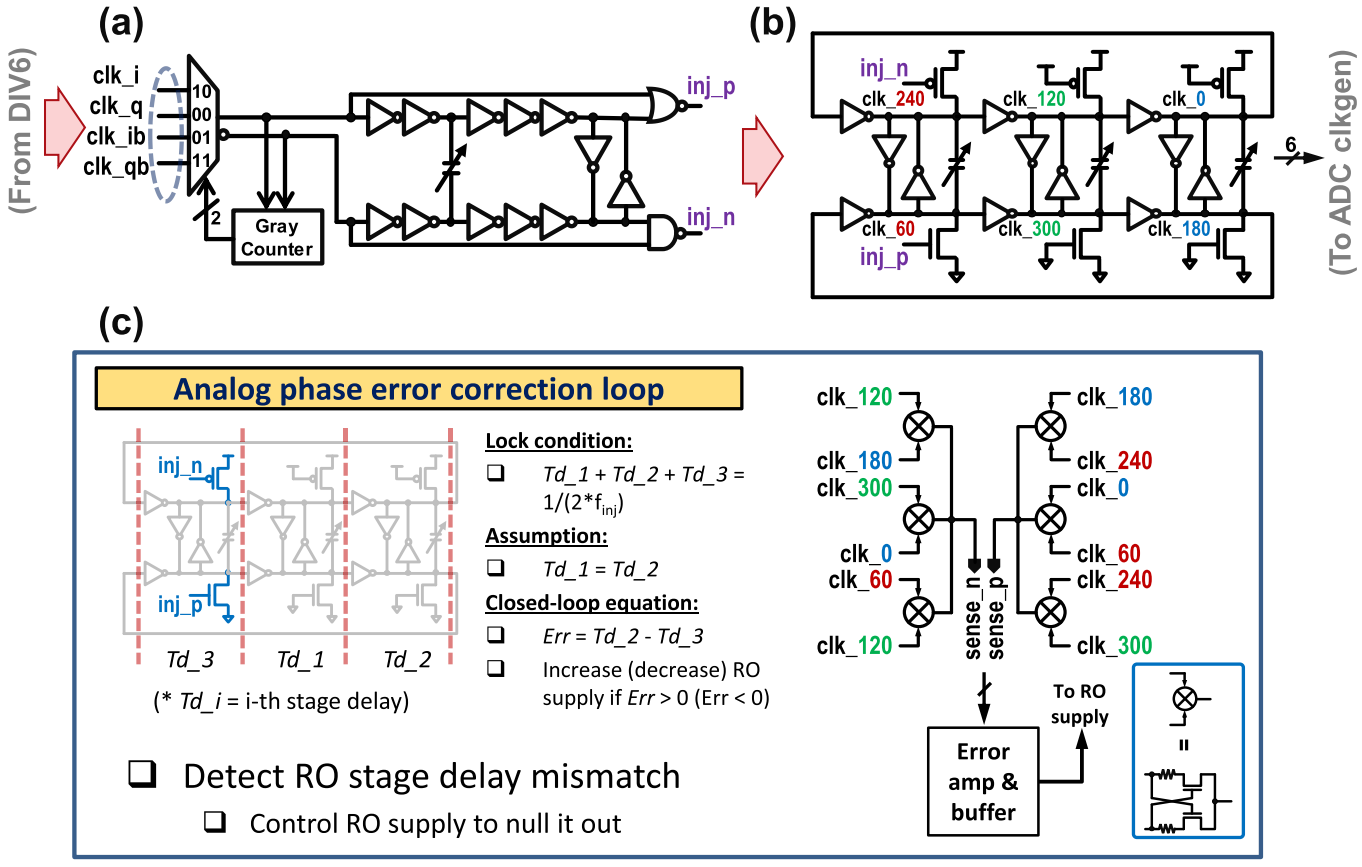


Fig. 10. Multiplying ILO circuits. (a) Pulse generator. (b) Injection-locked ring oscillator. (c) Phase error detector and correction loop.

9.333 GHz [Fig. 10(b)]. The combination of the multiplexer, pulse generator, and tuned ring oscillator together forms a multiplying-ILO (MILO) [11].

Fine control of the natural oscillation frequency of the MILO is required to achieve low phase spacing error at the six-phase output. The supply to the MILO is regulated by

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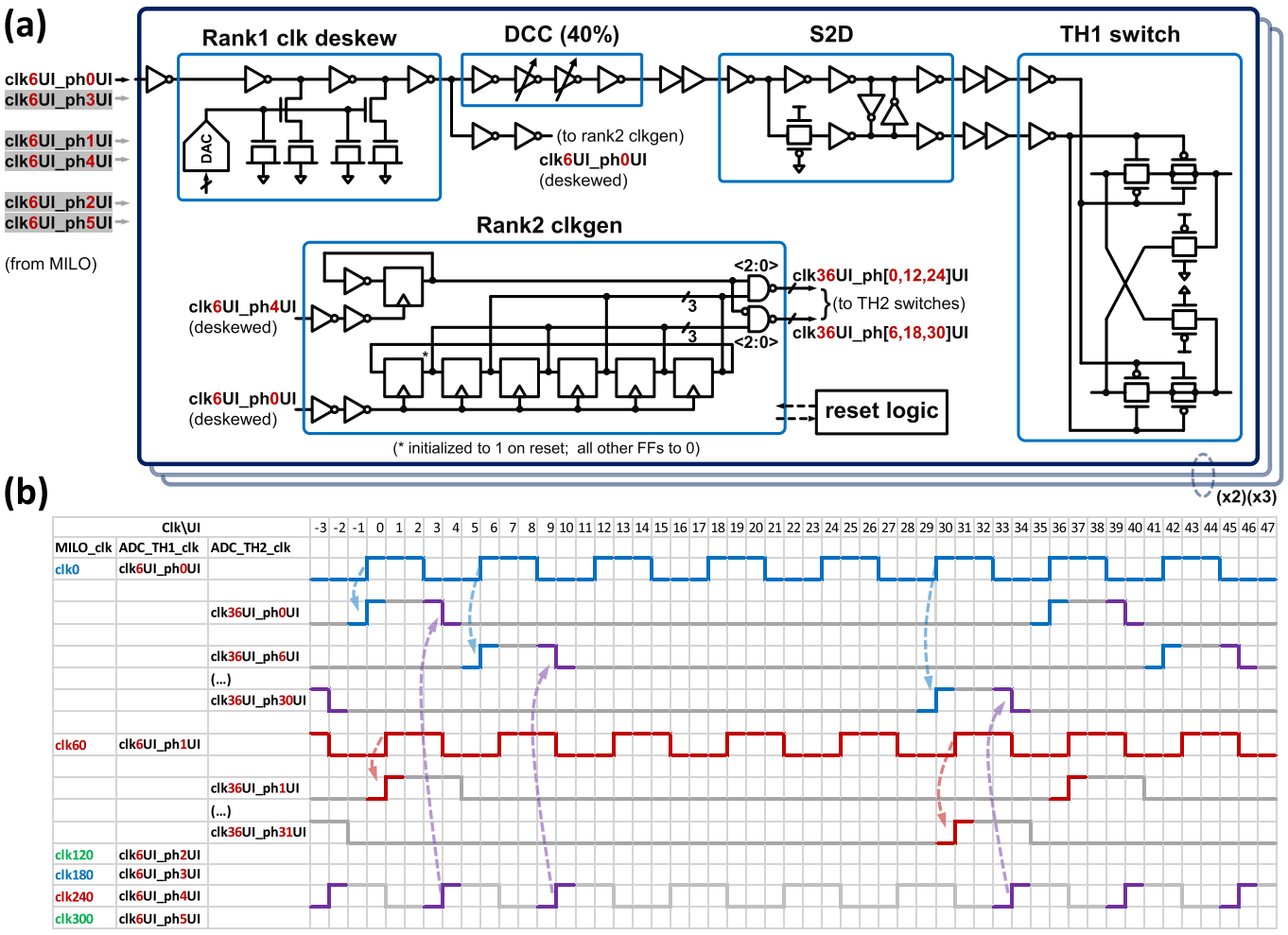


Fig. 11. ADC clock generation for rank 1 and rank 2 clocks. (a) Circuit implementation. (b) Timing diagram.

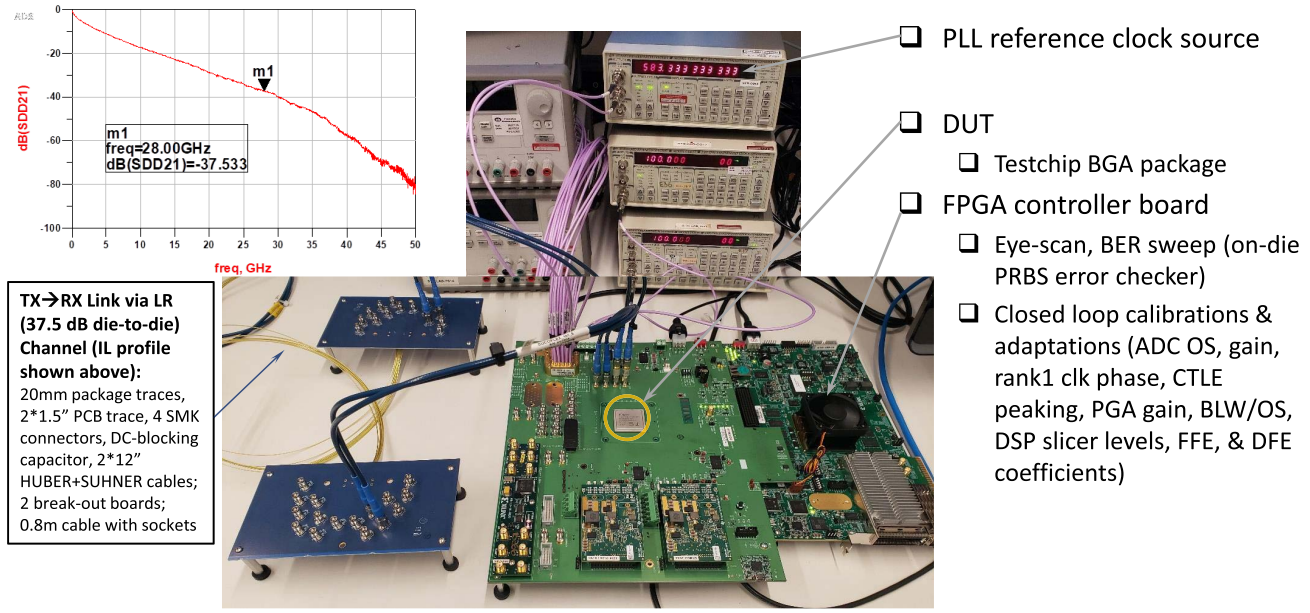


Fig. 12. Bench setup shown with TX to RX long-reach link performance test.

combining the six output phases to produce an error signal which is then integrated and filtered as illustrated in Fig. 10(c). The regulation has the effect of adjusting the natural oscillation frequency of the ILO such that its output period matches the input period. It is important to minimize the offset errors at the phase combination and error amplifier blocks as not all errors

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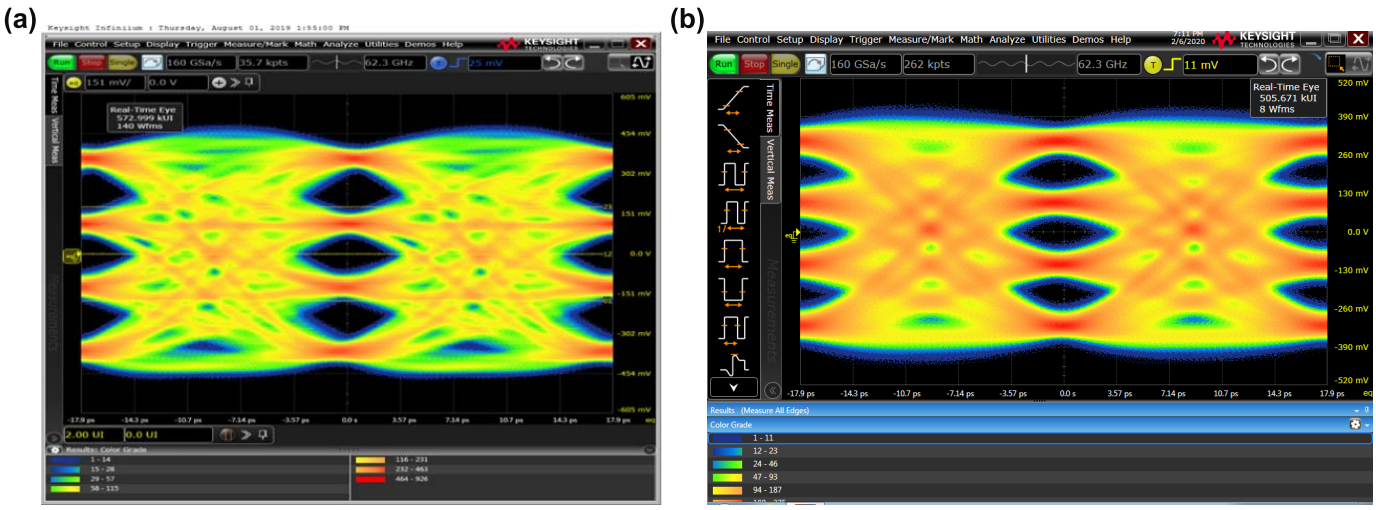


Fig. 13. 112-Gb/s PAM-4 TX output eye diagrams of (a) PRBS-7 and (b) PRBS-31 patterns.

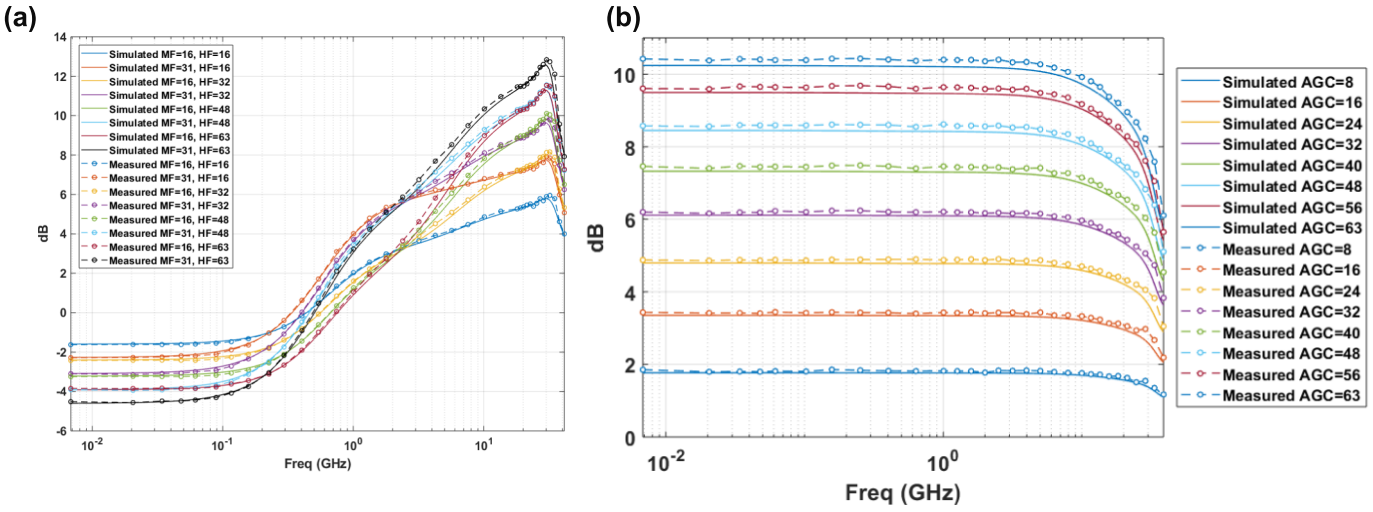


Fig. 14. RX AFE transfer function measurement versus simulation for (a) CTLE and (b) PGA.

accumulated are static or correctable. With a single control point, the average phase error at the MILO is minimized, but care is taken with the physical design to equalize the loading on the ring oscillator nodes to reduce the residual mismatch-induced errors.

The MILO method has the advantage that multi-phase clocks can be generated close to the locality of final use, lowering the distribution power. It should be noted for jitter purposes, however, that while the injected phases can have low-frequency jitter approaching that of the reference clock, high-frequency jitter (beyond the ILO bandwidth) will tend toward that of the intrinsic ring oscillator, and the non-injected phases will have further added jitter. Additional power-supply-induced jitter also requires careful budgeting.

It can be noted that instances of the pulse generation circuit being driven directly by the 2.333-GHz clocks would also produce a series of useful pulses, but that in such a case, multiple injection devices would be required. Instead, the use of fewer but higher rate injection devices lowers the loading on the ring oscillator and reduces multi-path mismatch.

While the MILO does have inherent suppression of timing errors at the injection point, residual offsets and random variations in both the MILO itself and downstream processing require further calibration. As such, the clocks are corrected for both skew and duty cycle, prior to application to the T/H switches in rank 1 or rank 2 ADC time interleavers.

The ADC clock generation circuit in Fig. 11(a) receives 6UI period six phase clocks from the MILO and processes them to produce T/H-1 and T/H-2 switching clocks. The skew correction is carried out using a DAC to adjust an MOS-based load on the clock paths. This affects both the rising and falling clock edges. The skew-adjusted clocks with near 50% duty cycle are divided for use in the rank 2 clocks, while the rank 1 clocks undergo programmable DCD adjustment to achieve a 40% duty cycle (2.4 UI tracking window) at 56GS/s ADC sampling speed. For power efficiency, the skew correction and DCD are applied to the six phases prior to a set of single to differential stages driving the T&H switches. T/H-2 clocks are generated by dividing the skew-corrected 50% duty cycle T/H-1 clocks by 6 to generate six-phase 6UI-long pulses

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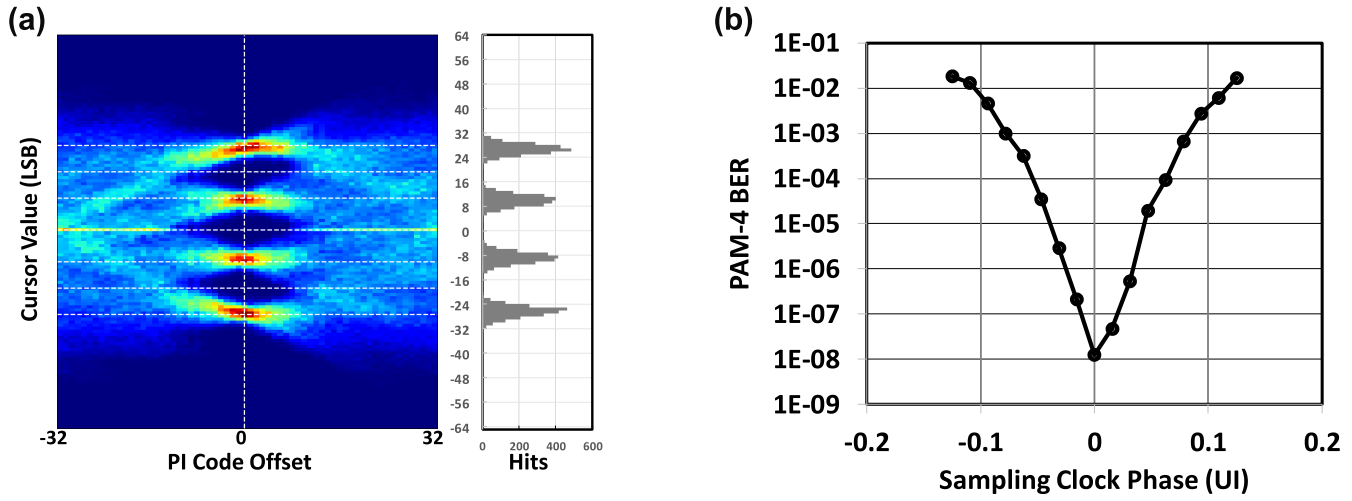


Fig. 15. Receiver performance for 112-Gb/s PRBS-31 over a long-reach channel (37.5 dB). (a) RX eye and cursor histogram. (b) BER bathtub curve.

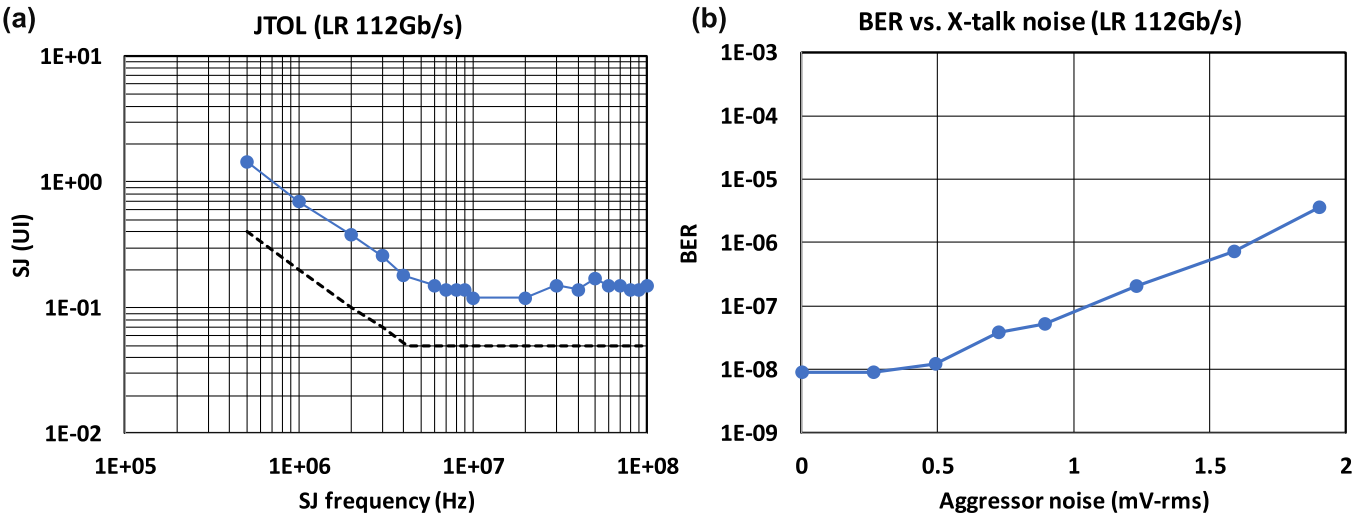


Fig. 16. (a) Sinusoidal jitter tolerance. (b) Cross-talk impact test for 112-Gb/s PRBS-31 over an LR channel.

of 36UI period. To meet the non-overlap timing condition, T/H-2 clocks are then trimmed by 2UI to have a 4UI tracking window as illustrated in the timing diagram in Fig. 11(b).

V. Bench Test

The fabricated transceiver wafer dice are assembled in ball grid array (BGA) packages, following a standard mass production VLSI packaging flow and are tested in a socket mounted on the device under test (DUT) PCB board. A bench test setup for TX-to-RX link performance over a long-reach channel is illustrated in Fig. 12. The DUT board with a DUT package socket and breakout cable connectors is controlled by another general-purpose FPGA board that fetches data and status bits from DUT and sends commands back to DUT. A control software performing PAM-4 RX adaptation and closed-loop corrections is coded and programmed onto the FPGA. An on-chip PRBS checker monitors the DSP data output to measure PAM-4 BER of the sliced data without sub-sampling. Contiguous raw ADC outputs (8-K samples) are captured and stored in the on-chip controller memory upon a trigger signal from the off-chip FPGA controller board. The FPGA controller periodically fetches the stored ADC output samples to perform continuous adaptation of data/error slice levels, FFE/DFE coefficients, CTLE peaking codes, PGA gain codes, AFE offset and baseline wandering correction DAC code, and ADC calibration codes (offset, gain, and skew). An external reference clock generator drives the on-die LC-PLL in DUT.

Fig. 13(a) and (b) shows the 112-Gb/s PAM-4 TX eye diagrams for the PRBS-7 and PRBS-31 patterns, respectively, using the built-in FFE of the Keysight DSA-X 96204Q oscilloscope. The TX output goes through 2 in of Megtron-6 PCB trace and a 24-in-long HUBER+SUHNER cable into the oscilloscope. PRBS-31 eyes show more symmetric PAM-4 eyes, albeit with reduced eye openings. The maximum TX swing is 1-V differential peak-to-peak and the measured random jitter is 140-fs rms for a clock pattern.

Fig. 14(a) and (b) plots the comparison between the measured and simulated composite transfer functions (TFs) of CTLE stages and PGA stages, respectively. The dashed lines

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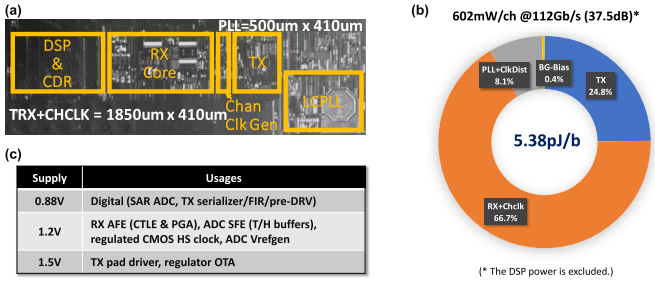
with markers represent the silicon-measured TFs for various peaking or gain code combinations. The solid lines represent the corresponding simulations. The TF mismatch between the measured data and simulation results is less than 0.5 dB across dc to 30 GHz, supporting the accuracy of our simulation methodology (e.g., parasitic extraction and field solver settings) for inverter-based CTLE and PGA circuits over a broad frequency range. The measured CTLE TF shows a max peaking of 17.5 dB at 30 GHz and its low-frequency zero is located near 300 MHz for the settings used. The measured PGA gain tuning range of 10.5 dB is observed.

To test the link performance, the TX output from the DUT board is looped back to the RX input via a channel consisting of TX/RX package traces (>20 mm total length embedded in an organic laminate substrate), two 1.5-in PCB traces, four SMK connectors, a dc-blocking cap, a pair of 12-in HUBER+SUHNER cables, two break-out boards, and a 0.8-m cable (Fig. 12). The total IL from TX to RX at 28 GHz is measured at 37.5-dB die-to-die, or 33.5-dB BGA-to-BGA excluding device package and DUT board socket, as illustrated in the SDD21 profile in Fig. 12. The measured transceiver performance is summarized as shown in Fig. 15(a) and (b). The transmitted 112-Gb/s PRBS-31 PAM-4 signal over the channel is effectively equalized by the RX at a BER <= 1E-8, as reported by the on-chip PRBS checker. The TX is configured to send a 930-mVppd signal with ~2-dB pre-cursor and ~5-dB post-cursor FFE boost.

The receiver eye in Fig 15(a) is plotted by accumulating 8-K symbol values of the equalized data slicer input in DSP while sweeping the sampling phase by overriding the PI code register by −32 to +32 offset around the CDR lock position. The corresponding cursor histogram is a vertical (ADC values measured in the unit of LSB) cross section of the RX eye taken at the PI code offset set to zero (i.e., at the CDR locking point), clearly showing equalized PAM-4 symbol levels. The measured horizontal BER bathtub in Fig. 15(b) shows a margin of >0.1UI to satisfy the OIF CEI pre-forward error correction (FEC) spec [6] of 1E-4 around the CDR locking point, where the BER reaches 1E-8 or lower.

A sinusoidal jitter (SJ) tolerance (JTOL) test in Fig. 16(a) was performed by setting a bit error rate tester (BERT) to inject SJ of varying frequency and magnitude to the DUT receiver over an LR channel. The data show an eye opening margin larger than 0.1UI at high frequency, while satisfying the OIF CEI JTOL mask [6] shown in the dashed line with a good margin across the frequency range from 500 kHz to 100 MHz. The results demonstrate the effectiveness of ISI cancellation by CTLE, FFE, and DFE as well as the jitter filtering capability of the on-chip digital CDR loop.

The cross-talk (X-talk) performance of ultra-high-speed SERDES, such as 112-Gb/s PAM-4 transceiver as reported in this article, has become very important due to the stronger aggressor-to-victim coupling resulting from sharper transition edges and the reduced SNR margin in PAM-4 signaling. The silicon characterization data of X-talk performance of such links provide valuable information to assist and guide system-level link margin budgeting and component specification. For X-talk noise margin testing, an asynchronous 56-Gb/s NRZ



(c)

Supply	Usages
0.88V	Digital (SAR ADC, TX serializer/FIR/pre-DRV)
1.2V	RX AFE (CTLE & PGA), ADC SFE (T/H buffers), regulated CMOS HS clock, ADC Vrefgen
1.5V	TX pad driver, regulator OTA

Fig. 17. (a) Die micrograph. (b) Power break-down. (c) Supply usages.

	This work	Ref [1]	Ref [2][4]	Ref [5]	Ref [3]
Technology	7nm FinFET	10nm CMOS	16nm FinFET	14nm FinFET	10nm FinFET
Data Rate	112Gb/s	112Gb/s	112Gb/s	128Gb/s	112Gb/s
TX Architecture	Quarter-rate CML 4-Tap FFE	-	Quarter-rate SST 4-Tap FFE	Quarter-rate CML 3-Tap FFE	Quarter-rate CML 3-Tap FFE
RX Equalization	CTLE 31-Tap FFE 1-Tap DFE	CTLE 16-Tap FFE 1-Tap DFE	CTLE 31-Tap FFE 1-Tap DFE	-	-
Active Area (excl. DSP)	TX: 0.136mm² RX: 0.265mm²	RX: 0.281mm²	TX: 0.38mm² RX: 0.674mm²	TX: 0.048mm²	TX: 0.03mm²
TX Power (excl. PLL)	1.34pJ/b	-	3.08pJ/b	1.0pJ/b (112Gb/s)	1.72pJ/b
RX Power (excl. DSP)	ADC: 4.8pJ/S RX: 3.6pJ/b*	ADC: 6.71pJ/S RX: 4.2pJ/b	ADC: 8.48pJ/S RX: 5.27pJ/b	-	-
PLL and Clk Distribution	0.44pJ/b	-	1.47pJ/b	-	-
Supplies	0.88V/1.2V/1.5V	0.95V/1.5V	0.9V/1.2V/1.8V	0.85V/0.9V	1.0V
Channel	37.5dB	35dB (BERT)	20dB (BERT)	-	-
BER (PRBS31)	1E-8	1E-6	2E-5	-	-

Fig. 18. Performance summary and comparison to prior published 112-Gb/s PAM-4 RX and TX with integrated PLL.

signal of varying amplitude from a BERT is injected into a neighboring channel as an aggressor. The total integrated noise power (in mV-rms) applied at the victim DUT RX input is first calibrated using a scope to measure the RMS difference of noise amplitude with and without the aggressor turned on. The IL between the scope and the X-talk test points was independently measured and accounted for. Plotted in Fig. 16(b), the BER degrades from ~1E-8 to ~1E-5 as the aggressor noise increases from 0- to 2-mV rms, leaving 10× margin from the OIF CEI pre-FEC BER spec of 1E-4.

Fig. 17(a) shows a die micrograph of the fabricated chip obtained by the back-side infrared imaging technique with major sub-blocks of the transceiver and the shared LC-PLL demarked by boxes. The measured power of 602 mW per transceiver channel for an LR channel shown in Fig. 17(b) consists of 25% of TX, 66% of RX, and 8% of amortized PLL and high-speed clock distribution contributions. A sub-ADC operating at 1.56 GS/s consumes 2.72 mW of active power, including T/H-2 clock buffers, a StrongArm slicer, SAR logic, CDAC, reference voltage, and ADC output retimers. Three power domains are used throughout the transceiver: 0.88 V for CMOS digital blocks; 1.2 V for RX AFE and ADC SFE stages and regulated supplies for LC-PLL, CMOS clock buffers, and ADC reference voltage; and 1.5 V for regulator error amplifiers and CML TX pad driver front-end.

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VI. Conclusion

Fig. 18 summarizes the transceiver performance and its comparison with prior published 112-Gb/s transmitters and receivers. The entire TX and RX dissipate 1.34 and 3.6 pJ/b, respectively, at 112 Gb/s over a long-reach channel, excluding DSP but including shared channel clock blocks. PLL and clock distribution buffers add 0.44 pJ/b per channel. The active TX and RX (excluding DSP and CDR) area are 0.136 and 0.265 mm^2, respectively.

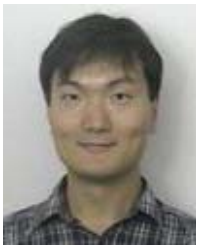
Compared with our previously reported 16-nm 112-Gb/s TX and RX [2], [4], this work achieved 60% of area and 40% of power reduction, thanks to the area-efficient all CMOS receiver front-end and clock generation and distribution circuits as well as a compact 36-way time-interleaving ADC SFE architecture. The bench test results demonstrated a superior performance over long-reach channels, satisfying the BER and JTOL requirements of OIF CEI-112G-LR-PAM4 standard [6].

Acknowledgment

The authors thank Xilinx SERDES design, layout, and verification teams for having made this project successful.

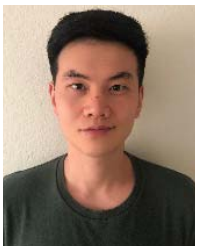
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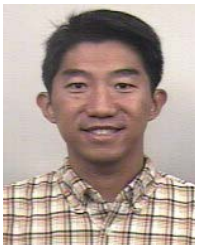
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Behzad Razavi

The StrongARM Latch

The StrongARM latch topology finds wide usage as a sense amplifier, a comparator, or simply a robust latch with high sensitivity. The term "StrongARM" commemorates the use of this circuit in Digital Equipment Corporation's StrongARM microprocessor [1], but the basic structure was originally introduced by Toshiba's Kobayashi et al. [2]. The StrongARM latch has become popular for three reasons: 1) it consumes zero static power, 2) it directly produces rail-to-rail outputs, and 3) its input-referred offset arises from primarily *one* differential pair. In this column, we study the circuit and its properties.

Basic Operation

Figure 1(a) shows the original StrongARM latch, reported in [2] without M_8 and in [1] with M_8. The circuit was later modified as depicted in Figure 1(b) [3]. We first study the latter and then point out the differences among these versions.

The latch of Figure 1(b) consists of a clocked differential pair, M_1–M_2, two cross-coupled pairs, M_3–M_4 and M_5–M_6, and four precharge switches, S_1–S_4. The circuit provides rail-to-rail outputs at X and Y in response to the polarity of $V_{in1} - V_{in2}$. We describe the operation in four phases.

In the first phase, CK is low; M_1 and M_2 are off; nodes P, Q, X, and Y are precharged to V_{DD}; and the circuit reduces to that shown in Figure 2(a).

In the second phase, CK goes high, S_1–S_4 turn off, and M_1 and M_2 turn on, drawing a differential current in proportion to $V_{in1} - V_{in2}$. With M_3–M_6 initially off, this current flows from C_P and C_Q [Figure 2(b)], thereby allowing $|V_P - V_Q|$ to grow and possibly exceed $|V_{in1} - V_{in2}|$. That is, this phase can provide voltage gain. We call this phase the amplification mode. Since the tail current is fairly constant during this period, we can write $|V_P - V_Q| \approx (g_{m1,2}|V_{in1} - V_{in2}|/C_{P,Q})\,t$, where $g_{m1,2}$ denotes the small-signal transconductance of M_1 and M_2, and $C_{P,Q} = C_P = C_Q$.

As V_P and V_Q fall to $V_{DD} - V_{THN}$, the cross-coupled NMOS transistors turn on (third phase), allowing part of the drain currents of M_1 and M_2 to flow from X and Y [Figure 2(c)]. The amplification mode therefore lasts for approximately $(C_{P,Q}/I_{CM})\,V_{THN}$ seconds, where I_{CM} is the common-mode (CM) current drawn from each capacitance. The voltage gain in this mode is roughly given by [4]

$$A_v \approx \frac{g_{m1,2}\,V_{THN}}{I_{CM}}. \qquad (1)$$

The behavior of the latch in the third phase can be analyzed with the aid of the equivalent circuit shown

Digital Object Identifier 10.1109/MSSC.2015.2418155
Date of publication: 25 June 2015

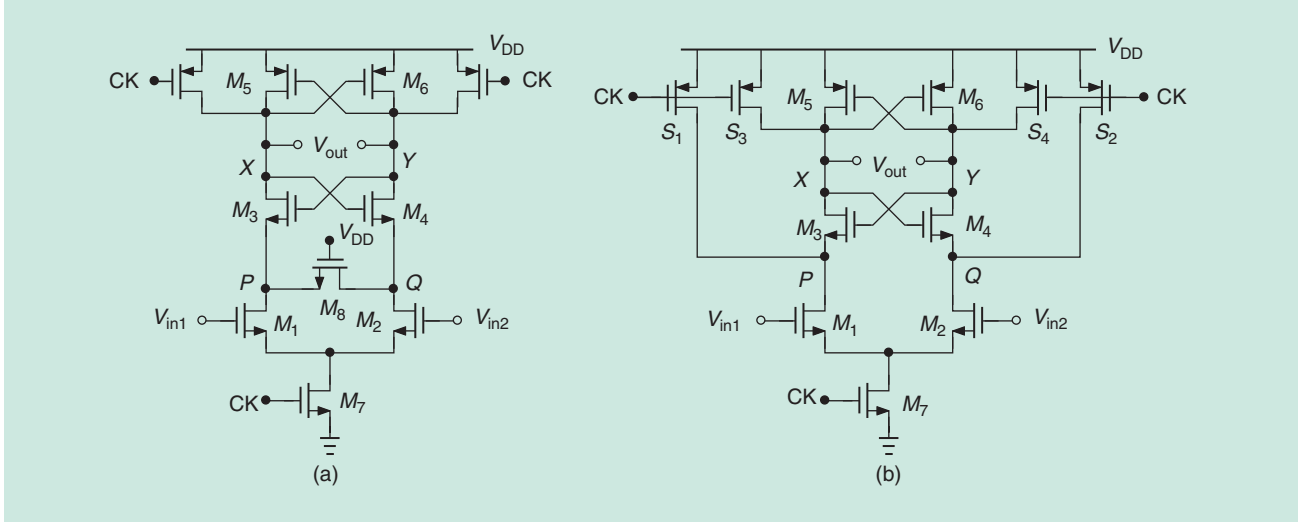


FIGURE 1: (a) The original and (b) modified StrongARM latch topologies.

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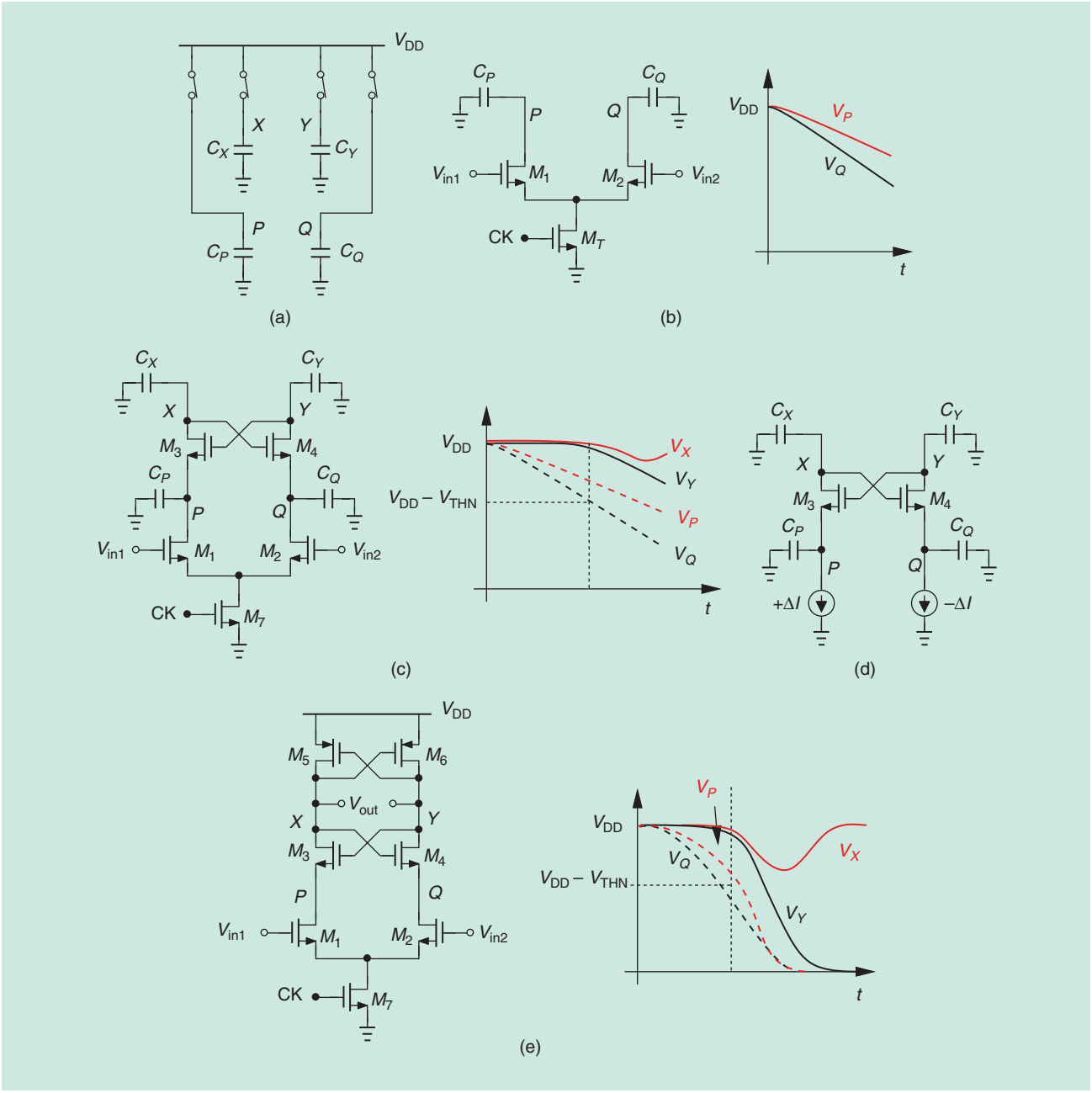


FIGURE 2: Latch operation phases: (a) precharge, (b) amplification, (c) turn-on of cross-coupled NMOS pair, (d) equivalent circuit of (c), and (e) turn-on of cross-coupled PMOS pair.

in Figure 2(d), where $+\Delta I$ and $-\Delta I$ represent the differential current produced by M_1 and M_2. Summing currents at the four nodes yields

$$-C_X\frac{dV_X}{dt} = g_{m3}(V_Y - V_P) \tag{2}$$

$$-C_Y\frac{dV_Y}{dt} = g_{m4}(V_X - V_Q) \tag{3}$$

$$-C_P\frac{dV_P}{dt} = C_X\frac{dV_X}{dt} + \Delta I \tag{4}$$

$$-C_Q\frac{dV_Q}{dt} = C_Y\frac{dV_Y}{dt} - \Delta I. \tag{5}$$

We subtract the second equation from the first, obtaining

$$-C_{X,Y}\frac{d(V_X - V_Y)}{dT} = g_{m3,4}(-V_X + V_Y - V_P + V_Q). \tag{6}$$

Integrating both sides of (4) and (5) and combining the results, we have

$$C_{P,Q}(V_Q - V_P) = C_{X,Y}(V_X - V_Y) + 2\Delta It, \tag{7}$$

which, upon substitution in (6), gives

$$C_{X,Y}\frac{d(V_X - V_Y)}{dt} - g_{m3,4}\left(1 - \frac{C_{X,Y}}{C_{P,Q}}\right)(V_X - V_Y) = -2g_{m3,4}\frac{\Delta I}{C_{P,Q}}t. \tag{8}$$

If $g_{m3,4}$ is assumed relatively constant, this equation reveals a natural response of the form $\exp(t/\tau_{reg})$, where τ_{reg} is the regeneration time constant and expressed as

$$\tau_{reg} = \frac{C_{X,Y}}{g_{m3,4}(1 - C_{X,Y}/C_{P,Q})}. \tag{9}$$

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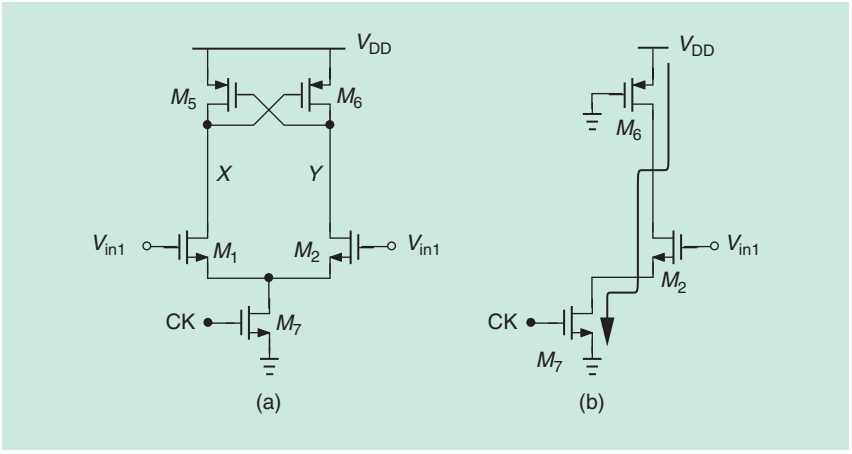


FIGURE 3: (a) Latch without ac cross-coupled NMOS pair and (b) the resulting static current.

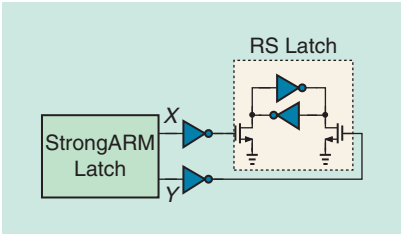


FIGURE 4: The StrongARM latch followed by the RS latch.

Interestingly, the degeneration caused by C_P and C_Q raises τ_{reg} by a factor of $1 - C_{X,Y}/C_{P,Q}$. Since, in practice, $C_{X,Y}$ includes the input capacitance of the stage following the comparator and is hence greater than $C_{P,Q}$, the cross-coupled NMOS transistors provide little regeneration in this phase.

The output voltages V_X and V_Y continue to fall until they reach $V_{DD} - |V_{THP}|$, at which point M_5 and M_6 turn on [Figure 1(e)] and the circuit enters the fourth phase. The positive feedback around these transistors eventually brings one output back to V_{DD} while allowing the other to fall to zero.

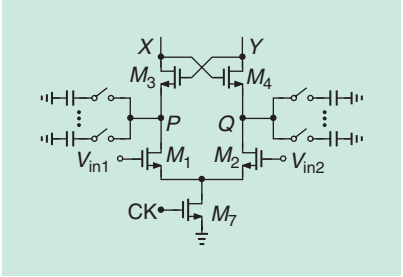


FIGURE 5: Offset cancellation by programmable capacitors.

It is important to appreciate the role of each transistor in the StrongARM latch of Figure 1(b). Besides M_1–M_2 and M_7, the remaining devices also serve critical purposes.

- Transistors M_3–M_4 cut off the dc path between V_{DD} and ground at the end of the fourth phase, avoiding static power drain. To understand this point, let us omit M_3 and M_4 as shown in Figure 3(a) and assume a differential input, $V_{in1} - V_{in2}$, of about 100 mV around a common-mode (CM) level near $V_{DD}/2$. When the latch is clocked, V_X falls, V_Y rises, and M_5 turns off. Consequently, the circuit reduces to that in Figure 3(b), drawing a static current from V_{DD}. (This does not occur for rail-to-rail inputs.)
- Transistors M_5 and M_6 principally restore the output high level to V_{DD}; without them, the CM discharge at X or Y would yield a degraded high level (if $|V_{in1} - V_{in2}|$ is small).
- Switches S_1 and S_2 play two roles: a) remove the previous states at nodes P and Q, suppressing dynamic offsets, and b) establish an initial voltage of V_{DD} at these nodes, allowing amplification before M_1 and M_2 enter the triode region. Both of these points distinguish the topology of Figure 1(b) from that in Figure 1(a). The original StrongARM latch fails to equalize V_P and V_Q accurately because M_8 turns off near the end of the precharge mode. Without M_8, the dynamic offset would prove even more serious. Moreover, the circuit has little voltage gain in the amplification mode for V_P and V_Q begin at $V_{DD} - V_{THN}$. Since in this case M_3 and M_4 turn on before significant gain accrues, they contribute a greater offset.
- Switches S_3 and S_4 precharge X and Y to V_{DD}, ensuring that M_5 and M_6 remain off during the initial amplification and negligibly raise the offset.

The StrongARM latch generates invalid outputs ($V_X = V_Y = V_{DD}$) for about half of the clock cycle. For the subsequent logic to interpret the outputs correctly, an RS latch must follow the circuit. Figure 4 shows a typical arrangement where inverters serve as buffers between the two latches and allow the RS latch to toggle only if V_X or V_Y falls.

The power consumed by the StrongARM latch of Figure 1(b) arises from primarily the charge and discharge of the capacitances. It is therefore roughly equal to $f_{CK}(2C_{P,Q} + C_{X,Y})V_{DD}^2$, where f_{CK} is the clock frequency and the factor of 2 accounts for the discharge of both P and Q to near ground in every cycle.

Offset

If operating as a sense amplifier or a comparator, the StrongARM latch must achieve a sufficiently small input-referred offset voltage. As explained in the previous section, the precharge action of S_1–S_4 in Figure 1(b) keeps M_3–M_6 off initially, thereby reducing their offset contribution. In a typical design, the mismatches between M_3 and M_4 are divided by about a factor of $A_v \approx 4$ when referred to the input, and those between M_5 and M_6 by about a factor of ten (because these transistors turn on only near the end). Thus, M_1 and M_2 become the dominant contributors.

Since the amplification mode provides voltage gain by the flow of charge from C_P and C_Q, one can create asymmetry by making $C_P \neq C_Q$ and hence cancel the circuit's offset. Illustrated in Figure 5 [5], [4], the idea is to establish different discharge rates at P and Q. Writing

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the drain current of each transistor as the sum of a component proportional to $V_{in1} - V_{in2}$ and a CM component, I_{CM}, we have

$$V_P = V_{DD} - \frac{g_{m1}(V_{in1} - V_{in2})}{2C_P} t - \frac{I_{CM}}{C_P} t \quad (10)$$

$$V_Q = V_{DD} + \frac{g_{m2}(V_{in1} - V_{in2})}{2C_Q} t - \frac{I_{CM}}{C_Q} t. \quad (11)$$

It follows that

$$V_P - V_Q = -\frac{g_{m1,2}}{2} \cdot \frac{C_P + C_Q}{C_P C_Q} (V_{in1} - V_{in2})\, t + \frac{C_P - C_Q}{C_P C_Q} I_{CM}\, t. \quad (12)$$

We observe that during amplification $V_P - V_Q$ accumulates an offset equal to $(C_P - C_Q)/(C_P C_Q) I_{CM} t$, which can cancel the latch's random offset. The amplification mode ends roughly when V_P and V_Q fall below $V_{DD} - V_{THN}$, and its duration is given by $t \approx V_{THN}(C_P + C_Q)/(2I_{CM})$, where $(C_P + C_Q)/2$ is used as an approximation. The built-in offset is therefore equal to $V_{THN}(C_P/C_Q - C_Q/C_P)/2$.

To perform offset cancellation, the main inputs are shorted together, the circuit is clocked, and the output decision drives a register that controls the values of C_P and C_Q [5], [4]. Of course, to reduce the offset from a high value (e.g., 30 mV) to a low value (e.g., 1 mV), a large number of small unit capacitors must be attached to P and Q, degrading the speed *and* raising the power dissipation. Another offset cancellation method for the StrongARM latch is described in [6].

Electronic Noise

From the foregoing offset studies, we can predict that the precharge action of S_1–S_4 in Figure 1(b) also reduces the electronic noise contributed by M_3–M_6. Most of the input-referred noise originates from M_1 and M_2 and the kT/C noise deposited by S_1 and S_2 because the other transistors come into play only after significant gain has accrued. In the amplification mode, the equivalent circuit of Figure 2(b) behaves as an integrator, generating output noise from the noise of M_1 and M_2. The variance of this voltage (a quantity akin to the mean square value) grows with time as [4], [7], [8]

$$E(V_{PQ}^2) = \frac{8kT\gamma g_{m1,2}}{C_{P,Q}^2} t. \quad (13)$$

Since the amplification mode lasts about $(C_{P,Q}/I_{CM}) V_{THN}$ seconds, we compute the final output noise variance due to M_1 and M_2 in this mode as

$$\sigma_{1,2}^2 = \frac{8kT\gamma}{C_{P,Q}} \cdot \frac{g_{m1,2} V_{THN}}{I_{CM}}. \quad (14)$$

Adding the kT/C noise contributed by S_1 and S_2, dividing the result by the square of the voltage gain, and writing $g_{m1,2} \approx 2I_{CM}/(V_{GS} - V_{THN})_{1,2}$, we obtain the total (integrated) input-referred noise observed in this mode as

$$\overline{V_{n,in}^2} = \frac{(V_{GS} - V_{THN})_{1,2}}{V_{THN}} \cdot \left[\frac{4kT\gamma}{C_{P,Q}} + \frac{(V_{GS} - V_{THN})_{1,2}}{V_{THN}} \frac{kT}{2C_{P,Q}} \right]. \quad (15)$$

The first term within the square brackets represents the noise due to M_1 and M_2 and is typically four to eight times greater than the second. Other sources of noise are quantified in [4].

While not specific to the StrongARM latch, the simulation of noise in comparators poses interesting issues. Unlike small-signal analog circuits, a comparator does not directly provide an output noise and a gain by which the noise should be divided. For simpler topologies, one can place the comparator in a metastable condition and perform a small-signal analysis, but the StrongARM latch completes switching actions and noise injections even *before* the output begins to change. A methodical simulation proceeds as follows. Suppose a comparator with a zero offset and a zero differential input is clocked many times (we assume the simulator includes noise in transient simulations). Then, the Gaussian noise within the circuit allows eventual recovery from metastability, producing ones and zeros at the output with equal probabilities [Figure 6(a)].

In the next step, we apply a small differential input (a few millivolts) as shown in Figure 6(b) and repeat the simulation. Since V_S skews the comparator decisions, the ones and zeros occur with unequal probabilities; zeros appear only if the input-referred noise is more negative than $-V_S$.

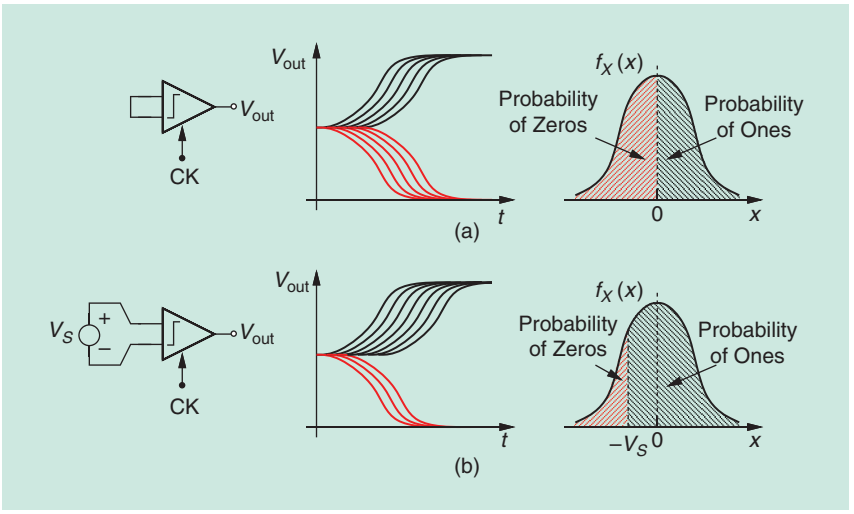


FIGURE 6: The behavior of noisy comparator with a (a) zero and (b) finite input differences.

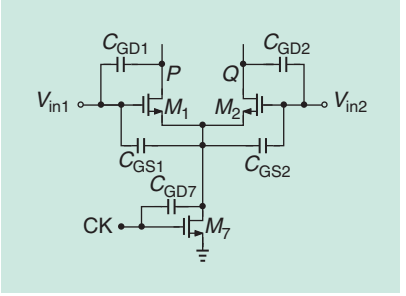


FIGURE 7: Kickback noise paths.

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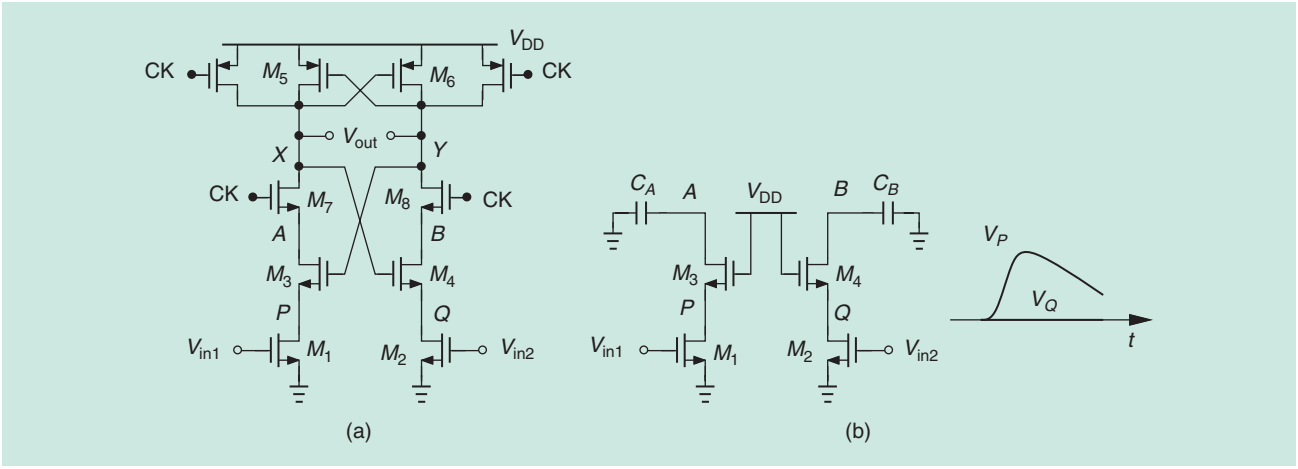


FIGURE 8: (a) An alternative topology for lower kickback noise and (b) behavior in the precharge mode.

For a large number of clock cycles, therefore, we predict that the number of zeros at the output, n_0, is proportional to the area under the Gaussian probability distribution function, $f_X(x)$, from $-\infty$ to $-V_S$; the number of ones, n_1, is proportional to the area from $-V_S$ to $+\infty$. Based on the numbers observed in the simulations, we can write

$$\frac{\int_{-\infty}^{-Vs} f_X(x)\,dx}{1-\int_{-\infty}^{-Vs} f_X(x)\,dx} = \frac{n_0}{n_1} \qquad (16)$$

and hence compute the variance of $f_X(x)$, which corresponds to the input-referred noise voltage squared. The value of V_S must be chosen large enough to ensure n_0/n_1 substantially departs from unity but not so large that n_0 or n_1 is excessively small and statistically insignificant.

Kickback and Supply Transients

The StrongARM latch draws high transient currents from the inputs and the supply. These transients become troublesome if a large number of comparators operate in parallel, as in a flash analog-to-digital converter.

The "kickback" currents drawn from the inputs stem from several mechanisms (Figure 7), exhibiting both differential and CM components. The former appear mostly as V_P and V_Q fall toward ground at unequal rates and couple to the inputs through C_{GD1} and C_{GD2}. This effect becomes more pronounced as M_1 and M_2 enter the triode region and their gate-drain capacitances increase. The CM kickback noise currents are much greater and occur when M_7 turns on, initially drawing its drain current from C_{GS1} and C_{GS2}, and when it turns off, with CK coupling through $C_{GD7}(\approx C_{GS7})$ to C_{GS1} and C_{GS2}.

The StrongARM latch draws high transient currents from the inputs and the supply.

It is possible to reduce the kickback noise by clocking the input devices through their drain path rather than their source path. Depicted in Figure 8(a) [9], such a topology incorporates M_7 and M_8 to control the latch. However, the kickback noise is lowered at the cost of a higher input offset because M_1 and M_2 now operate in the triode region during the amplification mode. This issue can be avoided by making M_3-M_4 and M_7-M_8 wide; but, as illustrated in Figure 8(b), the slow discharge at A or B in the precharge mode leads to significant imbalance between V_P and V_Q and hence a large dynamic offset.

The supply transient currents originate from the precharge action of S_1-S_4 in Figure 1(b). If CK falls fast, three of S_1-S_4 momentarily enter the saturation region (the fourth one is in the triode region because its drain voltage is equal to V_{DD}) and pull a large current from V_{DD}. The key point here is that designs consuming a low average power may still draw high peak currents from the supply, dictating a low supply impedance.

Questions for the Reader

1) Do V_P and V_Q in Figure 1(b) reach 0 V at the end of the regeneration phase?
2) Explain why M_3 and M_4 in Figure 1(b) can be omitted if the inputs have rail-to-rail swings.
3) Explain why the coupling through C_{GD7} in Figure 7 is less on the rising edge of CK than on the falling edge of CK.

You can share your thoughts with me by sending an e-mail to razavi@ee.ucla.edu.

Answers to Last Issue's Questions

1) Can we use a negative impedance converter (NIC) in a PA predriver to cancel the input capacitance of the output stage?

   Since an RF predriver typically uses a resonant load, the NIC would cause oscillation. If injection-locking is desired in this stage, a simple cross-coupled pair suffices.

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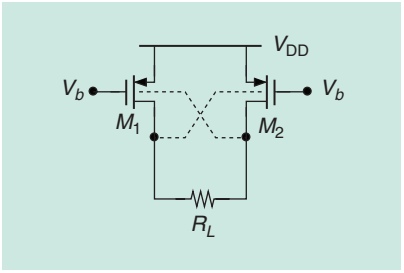


FIGURE 9: A cross-coupled pair using bulk terminals.

2) How does the thermal noise contributed by M_1 and M_2 in Figure 9 to V_{out} compare to that by a regular XCP?

This circuit produces a noise voltage across R_L equal to $\sqrt{2}\, g_m R_L / (2 - g_{mb} R_L V_n)$, where V_n denotes the gate-referred noise of each transistor and the noise of R_L is neglected. For a regular cross-coupled pair, the output noise is given by $\sqrt{2}\, g_m R_L / (2 - g_m R_L V_n)$. For a fair comparison, the total resistance seen at the output must be the same for the two topologies, thus yielding the same output noise.

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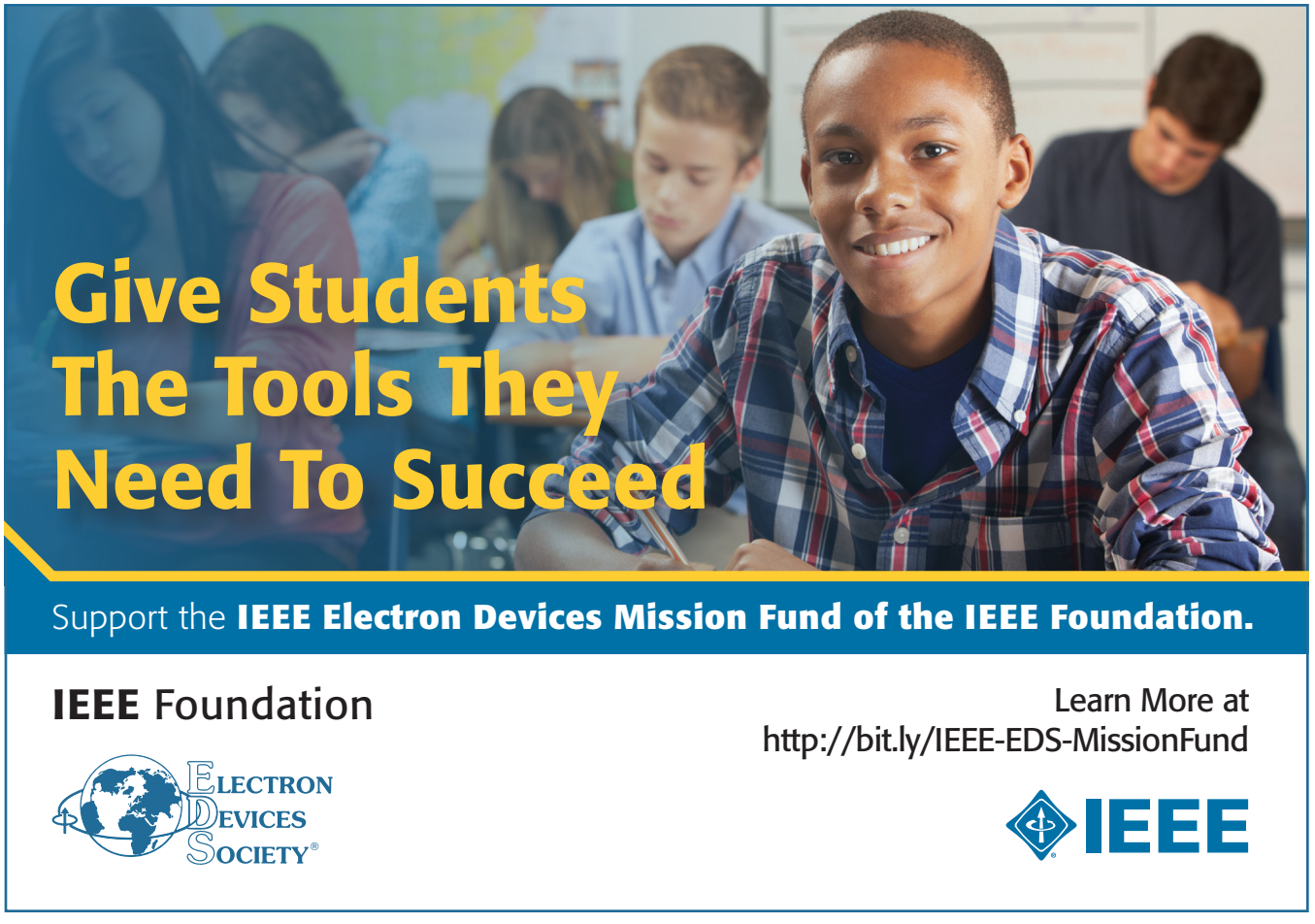
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A 9.1-ENOB 6-mW 10-Bit 500-MS/s Pipelined-SAR ADC With Current-Mode Residue Processing in 28-nm CMOS

Kyoung-Jun Moon, Dong-Shin Jo, Wan Kim, Michael Choi, Hyung-Jong Ko, and Seung-Tak Ryu, *Senior Member, IEEE*

***Abstract*— This paper introduces a current-mode residue processing technique in a pipelined-successive-approximation register (SAR) analog-to-digital converter (ADC), which extends the operation speed of a single-channel ADC utilizing low-impedance-based signaling. A 10-bit pipelined-SAR ADC with featured building blocks such as a degenerated gm-cell as an open-loop residue amplifier, a switched-current mirror for sample-and-hold (S/H) function, and a split current digital-to-analog converter (DAC) for current-domain SAR conversion achieves a 500-MS/s conversion-rate under a 1.0-V supply. With background inter-stage mismatch calibration, a prototype ADC fabricated in a 28-nm CMOS process achieves 56.6-dB signal-to-noise-and-distortion ratio (SNDR) at a Nyquist input, resulting in a Walden figure of merit (FoM) of a 21.7-fJ/conversion-step.**

***Index Terms*— Analog-to-digital converter (ADC), current-domain, current-mode residue processing, gm-cell, open-loop amplifier, pipelined-successive-approximation register (SAR), split current digital-to-analog converter (DAC).**

I. Introduction

RECENTLY reported pipelined-successive-approximation register (SAR) analog-to-digital converters (ADCs) combining moderate resolution sub-stage SAR ADCs with a residue amplifier have shown not only excellent power efficiency but also high conversion rates [1]–[10]. Although the conversion speed of ADCs with a pipelined-SAR architecture can be significantly enhanced with time-interleaving (TI) techniques [3]–[8], mismatches between sub-channels in TI ADCs present a considerable design bottleneck. With the channel mismatch problems in highly interleaved ADCs as motivation, some previous designs have been developed to enhance the single-channel speed of pipelined-SAR ADCs [9], [10]. Various architectural modifications of ADCs in their internal SAR ADCs have been proposed, such as multi-bit/cycle conversion [9] and a loop-unrolled structure [10]. Nevertheless, the residue amplifier required for pipelining has always been a major design challenge for speed, gain accuracy, and power consumption.

Design techniques that replace opamp-based residue amplifiers with simple open-loop amplifiers have been exploited for low power and high speed. As reported in [4]–[6] and [9], the integrated dynamic amplifiers can provide a very power-efficient solution, but the voltage-to-time conversion process makes the structure very sensitive to process–voltage–temperature (PVT) variation. The background gain calibration [6] and the replica-aided dynamic amplifier [9] schemes could enhance the robustness, but they accompany design complexity and additional power consumption. In addition, the dynamic amplifiers basically require an integration time period or high supply voltage for high gain and a high signal-to-noise ratio (SNR) [11], and this would limit the speed of the residue generation. This time and supply dependency problem of the dynamic amplifiers could be alleviated with a traditional open-loop voltage amplifier with a resistive load [10]. However, the *RC* time constant with the capacitive load of the subsequent SAR ADC can also limit the operation speed.

For a power-efficient high-speed single-channel pipelined-SAR ADC, a current-mode open-loop residue processing scheme [12] is introduced in this paper. Even though current-mode pipelined ADCs have been previously reported as [13]–[15], all of them utilized a closed-loop approach. The proposed ADC employs a gm-cell as an open-loop residue amplifier, and a high-speed current-steering digital-to-analog converter (DAC) is utilized for the fine SAR ADC for low-impedance-based high-speed signaling.

This paper is organized as follows. In Section II, the proposed current-mode open-loop residue processing scheme is discussed. The detailed circuit implementation and an analysis of the proposed pipelined-SAR ADC are then provided in Section III. The measurement results of the prototype ADC are presented in Section IV, followed by conclusions in Section V.

II. Proposed Current-Mode Open-Loop Residue Processing

A conventional pipelined-SAR ADC with a voltage-mode open-loop residue processing scheme is modeled in Fig. 1. It consists of an open-loop residue amplifier with a resistive load (R_L) and an N-bit coarse and M-bit fine SAR ADCs with a capacitive DAC (C-DAC). The output voltage (V_{OUT}) of the residue amplifier is obtained as

$$V_{\mathrm{OUT}}(t) = V_{\mathrm{RES}} \cdot G_{\mathrm{m}} \cdot R_L \cdot \left(1 - e^{-\frac{t}{R_L \cdot C_L}}\right). \quad (1)$$

Manuscript received December 14, 2018; revised March 19, 2019 and May 30, 2019; accepted June 27, 2019. Date of publication July 23, 2019; date of current version August 23, 2019. This paper was approved by Associate Editor Jeffrey Gealow. *(Corresponding author: Seung-Tak Ryu.)*

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Digital Object Identifier 10.1109/JSSC.2019.2926648

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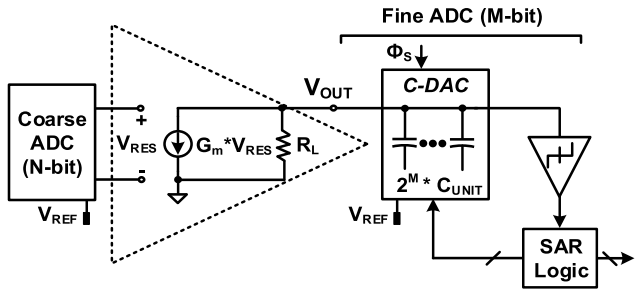


Fig. 1. Voltage-mode residue processing in pipelined-SAR ADC.

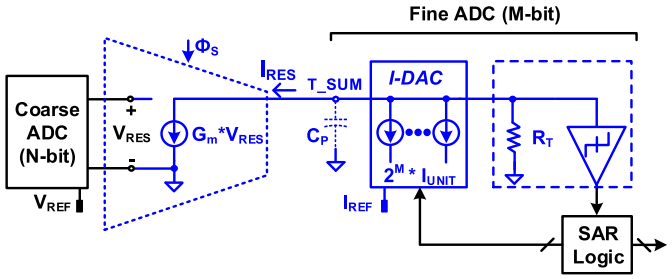


Fig. 2. Proposed current-mode residue processing in pipelined-SAR ADC.

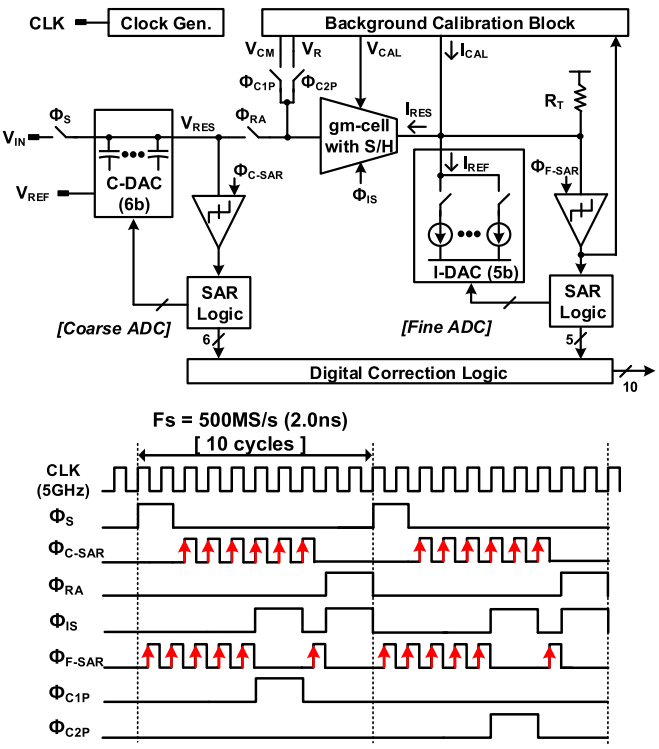


Fig. 3. Block and timing diagram of the 10-bit 500-MS/s pipelined-SAR ADC.

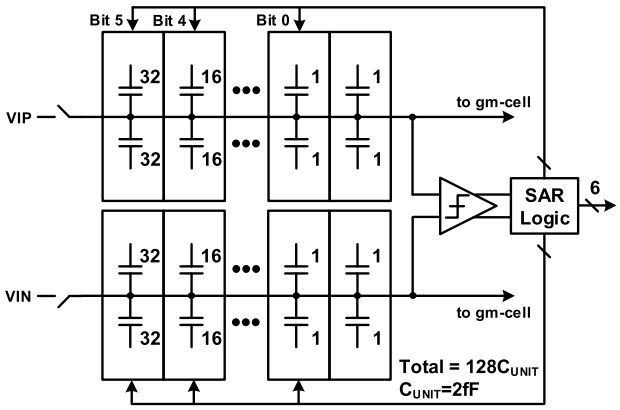


Fig. 4. Implementation of the 6-bit coarse ADC.

With a residue voltage (V_{RES}) generated from the coarse SAR ADC as the input of the amplifier, the operation speed of the single-pole open-loop amplifier is determined by the RC time constant (τ) at the output node of the amplifier. In particular, the gain of the residue amplifier ($\propto R_L$) and the total capacitance ($2^{\mathrm{M}} \cdot C_{\mathrm{UNIT}}$) of the C-DAC in the fine SAR ADC are the parameters that provide the largest contribution to the RC time constant.

In accordance with trends of recently advanced CMOS processes, the C-DAC in the fine SAR ADC can be implemented with a small C_{UNIT} of less than 1 fF, showing excellent matching performance [11]. Also, it is possible to minimize the load resistance (R_L) by reducing the residue gain of the amplifier, which further alleviates the non-linearity characteristic of the residue amplifier with a small output swing level. The total capacitance ($2^{\mathrm{M}} \cdot C_{\mathrm{UNIT}}$) of the C-DAC, however, directly affects the performance of the fine ADC, such as kT/C noise, comparator kickback noise, and signal attenuation due to the input parasitic capacitance of the comparator. Therefore, as the resolution of the fine ADC increases, a larger total capacitance is required for the C-DAC to realize noise-immune pipelining. In addition, the influence on the RC time constant by additional parasitic capacitances such as a junction and metal line capacitance at the output node of the amplifier is not negligible. Hence, voltage-mode open-loop residue processing with a C-DAC for fine SAR restricts the RC time constant minimization for high-speed operation.

By replacing the conventional C-DAC with a current-steering DAC (I-DAC), as depicted in Fig. 2, the proposed current-mode open-loop residue processing scheme alleviates the capacitive load, and thus, facilitates higher speed residue generation. The proposed pipelined-SAR ADC consists of a gm-cell with a sample-and-hold (S/H) function, an I-DAC with a reference current (I_{REF}), and a comparator with a trans-impedance resistor (R_{T}) for fine SAR conversion. The gm-cell converts the residue voltage (V_{RES}) to current (I_{RES}) and holds it by utilizing a sampling clock (Φ_{S}). The settling time required to amplify the residue through the gm-cell, similar to the voltage-mode pipelined-SAR ADC as (1), is decided by the parasitic capacitance (C_{P}) and load resistance (R_{T}) at the current summing node (T_SUM). Unlike the total capacitance (C_L) of the C-DAC in Fig. 1, the parasitic capacitance (C_{P}) is independent of the signal level ($\propto R_{\mathrm{T}}$) at the node of T_SUM determining the SNR of the fine ADC. By minimizing the C_{P}, a low RC time constant can, therefore, be realized without sacrificing the SNR performance. As the voltage swing level at the node of T_SUM becomes smaller, the I-DAC has more voltage headroom, and thus, the linearity and noise performance

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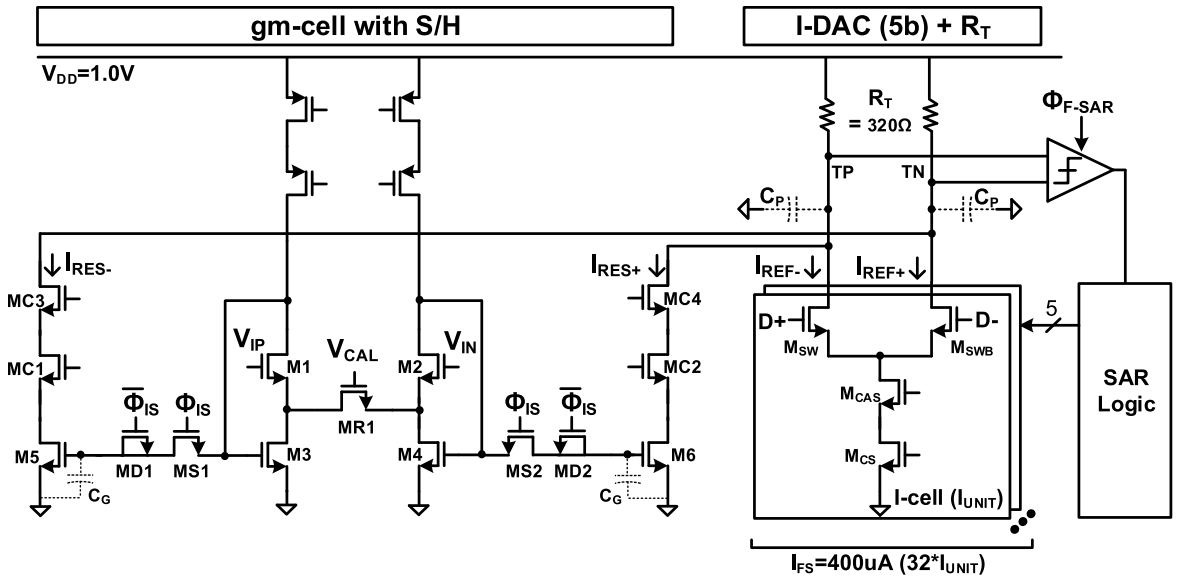


Fig. 5. Circuit Schematic of the gm-cell with S/H and the 5-bit fine SAR ADC.

of the fine ADC can be secured. Hence, the current-mode residue processing technique is more suitable for low supply. The parasitic capacitance (C_P) consists of the device junction capacitance and metal line parasitic of the I-DAC, and a very small value can be obtained by proper biasing in the unit current cell and a compact layout. The estimated C_P of the prototype ADC is below 30 fF. The proposed current-mode residue processing scheme results in low-impedance signaling for fast residue settling while realizing a noise-optimized design.

III. Circuit Implementation

A. Overall Architecture and Coarse ADC

Fig. 3 presents an overall block diagram of the proposed ADC with its operational timing diagram. The proposed ADC is composed of a 6-bit voltage-mode coarse SAR ADC, a gm-cell with current-mode S/H function for residue amplification, and a 5-bit fine SAR ADC with a current-steering DAC (I-DAC). With a 1-bit redundancy between the coarse and fine SAR ADCs, the entire ADC generates a total of 10-bit digital output. A background calibration function is embedded on a chip to calibrate the errors between the coarse and fine stages. The 10-bit A/D conversion is completed within ten internal clock cycles. The input voltage (V_{IN}) is sampled on the C-DAC for 1.5 clock cycles and 6-bit coarse A/D conversion is then conducted for the following 6.5 clock cycles. After the coarse A/D conversion is completed in a total of eight cycles, the residue voltage (V_{RES}) remaining on the C-DAC is converted into current (I_{RES}) by the gm-cell for the last two clock cycles. The residue current is sampled and held in the gm-cell by the S/H function at the end of the tenth clock cycle. During the first five clock cycles in which the coarse ADC samples the next input and conducts the SAR operation with it, the 5-bit fine ADC with its I-DAC conducts SAR operation with I_{RES}. While the coarse A/D conversion is still in process, during the three clock cycles after the fine SAR operation is completed, the gm-cell and the fine ADC conduct offset and gain calibration with the relevant calibration circuitry.

As shown in Fig. 4, a split binary-weighted a C-DAC with a top-plate sampling scheme is employed in the voltage-mode coarse SAR ADC [16]. This switching scheme enables high-speed DAC switching with simple SAR logic and maintains the common-mode voltage of the gm-cell constant regardless of the code. The total C-DAC capacitance of the 6-bit coarse SAR ADC is designed to be 256 fF considering the kT/C noise and capacitor mismatch for a 10-bit resolution with a 1.6-V full-scale differential swing. A total error of about 25 mV of the comparator offset and the DAC settling in the coarse ADC can be tolerated by the 1-bit inter-stage redundancy.

B. Current-Mode Back-End

Fig. 5 shows a detailed block diagram of the residue-amplifying gm-cell with S/H and the 5-bit fine SAR ADC. The gm-cell with S/H function is composed of a flipped voltage follower (FVF)-structured input stage (M1–M4) [17] with a degeneration resistor (MR1) for linear voltage-to-current conversion and a switched-current mirror (V–I) to hold the charges on its gate capacitance (C_G) for current sampling (M5, M6, MS1, MS2). For linearity of the voltage-to-current conversion gain, a triple cascode current mirror (M5, M6, MC1–MC4) and dummy switches (MD1, MD2) are utilized. The fine ADC consists of a 5-bit I-DAC, a voltage comparator with a trans-impedance resistor (R_T) and a SAR logic. The differential residue voltage of V_{IP} and V_{IN} are converted into currents, I_{RES-} and I_{RES+}, by the gm-cell. For the SAR operation, the residue current sampled by the clock Φ_{IS} is compared with the currents from the 5-bit I-DAC, I_{REF+}, and I_{REF-}. All current sources in the gm-cell and the I-DAC

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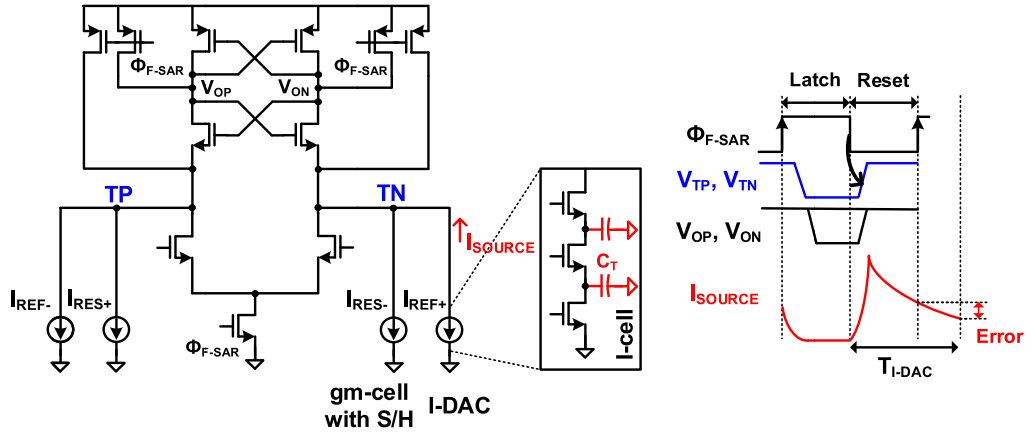


Fig. 6. SAR operation with current comparison.

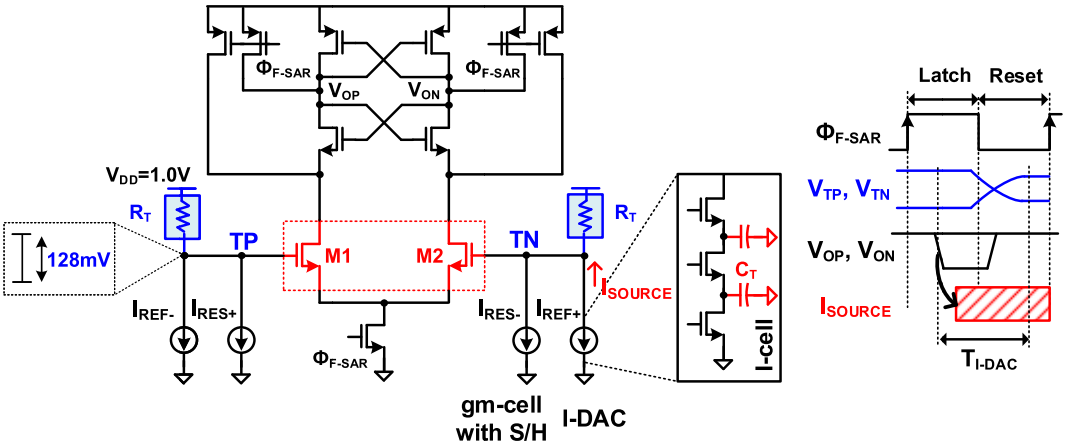


Fig. 7. SAR operation with voltage comparison with R_T.

employ only an NMOS transistor for a supply voltage under 1.0 V, enhancing the operation speed and mitigating design complexity.

1) Voltage Comparator With R_T: Since both the amplified residue and the DAC signal in the fine ADC are in current mode, it would be natural to choose a current comparator that takes the current signals directly as its input, as conceptually shown in Fig. 6. However, with this structure, a speed issue arises when the reset time of the comparator becomes short (i.e., in high speed operation) due to the slow DAC turn-on time: At the end of the latching phase, both V_{TP} and V_{TN} reach ground level, and thus, all current sources enter the triode region. During the following reset phase, V_{TP} and V_{TN} are reset to the supply level, and the current sources turn on for DAC operation by initially charging the parasitic capacitance (C_T) of the internal nodes. As the reset time becomes shorter, however, the current sources cannot be fully turned on due to the insufficient parasitic charging. This results in insufficient DAC settling, which leads to erroneous SAR conversion. In order to avoid this problem in the gm-cell and the I-DAC, a small resistive load (R_T) is connected to the current output, as shown in Fig. 7. The R_T converts the current output into voltage, and thus, a voltage comparator conducts the comparison. Therefore, the current sources are always turned on regardless of the reset operation of the comparator and the DAC settling time for the SAR operation can be ensured, which enables high-speed SAR operation of the fine ADC.

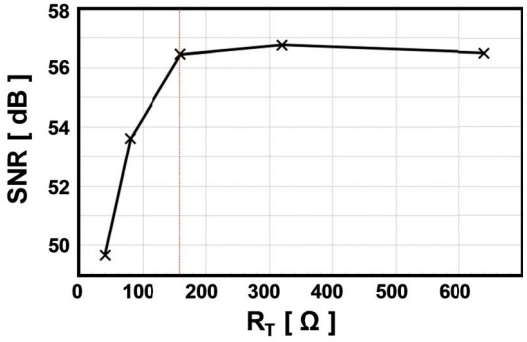


Fig. 8. Transient noise simulation result of ADC's SNR versus R_T.

Accordingly, the fine SAR ADC is constructed with a current DAC that drives the R_T and a strong-arm comparator. In terms of residue settling speed, a smaller R_T is preferred, and therefore, R_T is chosen to be as small as possible until the voltage swing level at the summing node (TP, TN) limits the SNR of the fine ADC. In this structure, the noise

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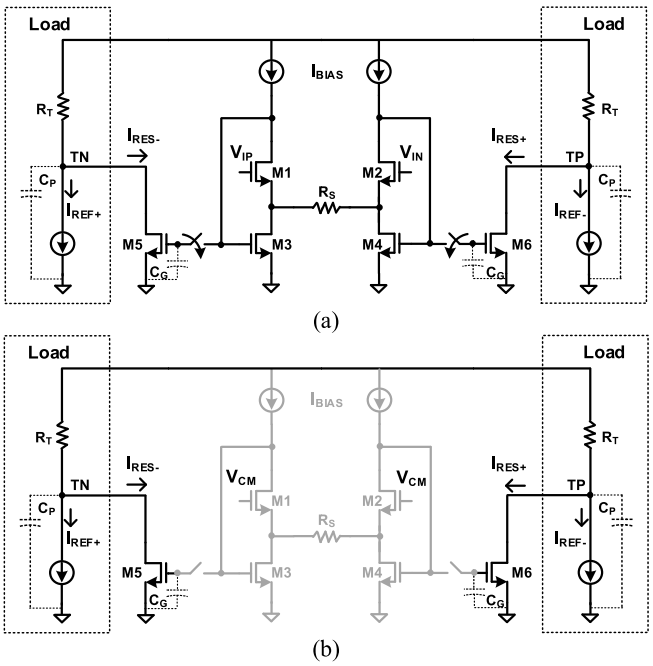


Fig. 9. Circuit schematic of the gm-cell (a) in the sampling phase and (b) in the holding phase.

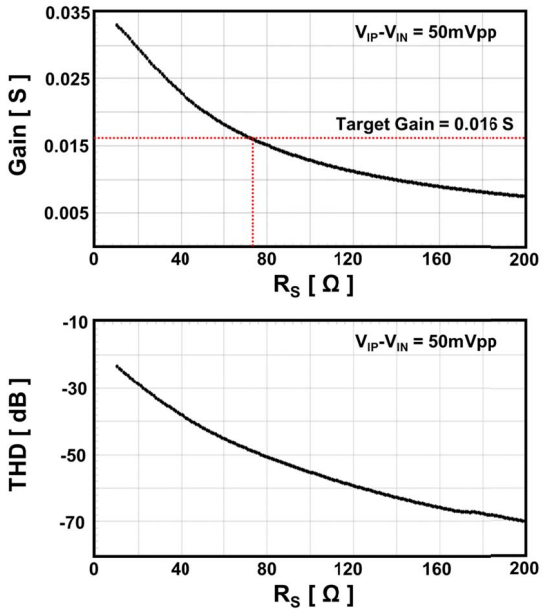


Fig. 10. Simulated gain and THD versus R_S in the gm-cell.

characteristics of the fine ADC are determined mainly by the thermal noise of the comparator. A smaller R_T at the node of TP and TN leads to a higher common-mode level at the input of the comparator, which causes a faster but noisy operation of the comparator with a normal threshold transistor. The effective overdrive voltage of the comparator input transistor (M1, M2) is, thus, reduced by utilizing a high threshold device to minimize the noise influence of the comparator.

Fig. 8 presents the transient noise simulation results showing the SNR performance of the overall 10-bit ADC according to the R_T value with the full-scale current of 400 μA of the I-DAC. The simulated minimum value of R_T at which the noise impact of the comparator in Fig. 7 does not degrade the overall ADC's SNR of about 57 dB (dominated by the noise of the quantization and the gm-cell) is roughly 160 Ω. Considering the process and temperature variations of the resistance (>20%), an R_T value of 320 Ω is, thus, utilized and the residue swing range at the summing node (TP, TN) is around 128 mV.

2) gm-Cell With S/H: Fig. 9 depicts the operational principle of the gm-cell in (a) the sampling phase and (b) the holding phase. In the sampling phase, the small-signal voltage-to-current conversion gain of the gm-cell is

$$\frac{I_{RES+} - I_{RES-}}{V_{IP} - V_{IN}} \approx \frac{1}{\frac{1}{g_{m1} \cdot g_{m3} \cdot r_{o1}} + \frac{R_S}{2}} \tag{2}$$

where g_{m1} and g_{m3} are the trans-conductance of M1 and M3, respectively, and r_{o1} is the drain–source resistance of M1. The simulation results of the gm-cell gain and the total harmonic distortion (THD) according to the degeneration resistor value, R_S, are shown in Fig. 10. With R_S and the negative feedback characteristic of the FVF, the gm-cell obtains a well-defined voltage-to-current gain while securing linearity. In this design, the maximum input voltage of the gm-cell as the residue from the coarse ADC is about 50 mV. With the R_T value of 320 Ω and voltage swing at the output (TP, TN) of around 128 mV, as mentioned above, the full-scale output current of the gm-cell (I_{RES+}, I_{RES-}) is differential 800 μA. The gm-cell with the target voltage-to-current gain of 0.016 S should satisfy the 5-bit linearity. Hence, R_S is designed to be about 80 Ω with a −47-dB THD. With a THD drop of less than 1 dB, an NMOS transistor in the triode region, the MR1 shown in Fig. 5, realizes R_S for gain calibration purpose (PVT-insensitivity of the open-loop amplifier) with a controlled gate voltage of V_{CAL}. While the gm-cell holds the sampled residue current owing to the gate capacitance (C_G) of the mirror transistor, as shown in Fig. 9(b), the gate capacitance of the input transistor of the FVF stage should be reset to remove the memory effect. Although the input nodes of the gm-cell can be reset to the ground level for power savings during the holding phase, as the recovery time from reset to the subsequent sampling phase can degrade the settling time, a V_{CM} of around 0.55 V is utilized for input reset. This reset operation is also required for the calibration mode, as will be discussed in detail in Section III-D.

3) Split-Current Source DAC: The current DAC structure and its switching algorithm have been modified in this work to improve the DAC settling speed of the fine SAR ADC. A representative conventional switching technique of a 5-bit binary-weighted current-steering DAC is shown in Fig. 11(a). For MSB conversion, a differential zero signal (IREF_diff = 0) is generated by connecting the MSB current source of $16I_{UNIT}$ to one output (e.g., I_{REF-}) while the remaining current sources, whose total current is also $16I_{UNIT}$, are connected to the other output (I_{REF+}). When the MSB is determined to be one by the SAR operation, the MSB-1 current source of $8I_{UNIT}$ is connected to the negative output, and the $16I_{UNIT}$ is switched back to the positive output for the following MSB-1 conversion.

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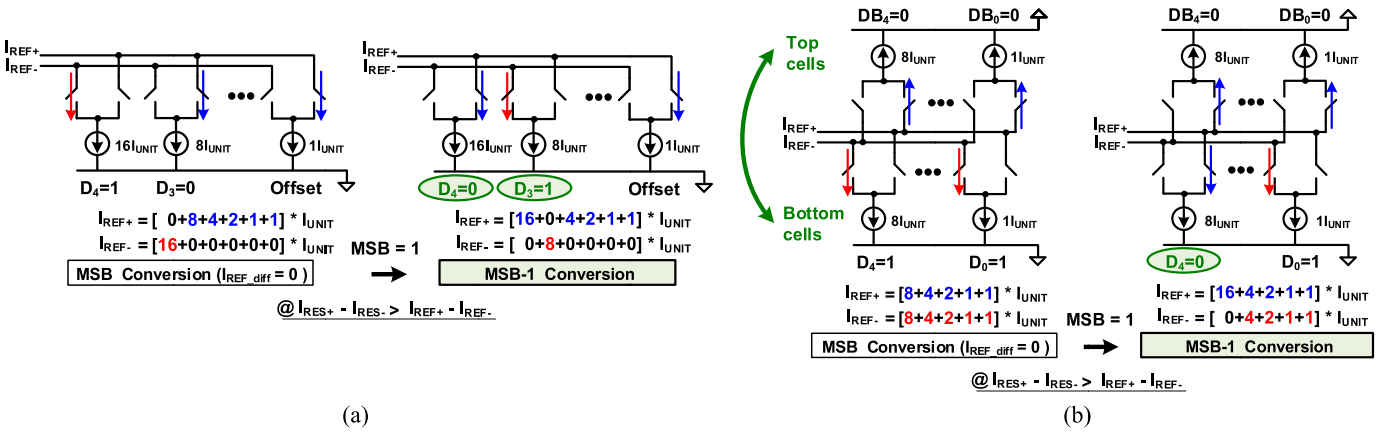


Fig. 11. Switching algorithm of the (a) conventional current-steering DAC and (b) proposed split-current source DAC.

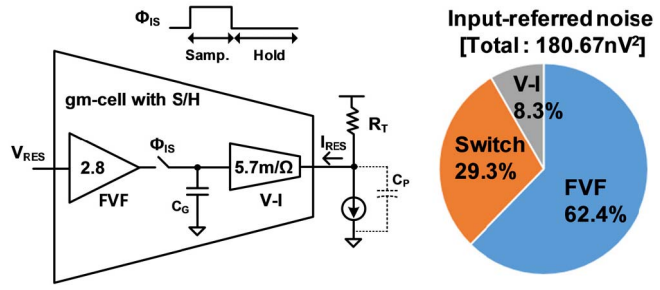


Fig. 12. Noise analysis of the gm-cell with S/H.

TABLE I
SIMULATED NOISE CONTRIBUTION

	Noise [nV²]	[%]
Quantization	203.45	37.7
Coarse ADC	101.1	18.7
gm-cell with S/H	180.67	33.5
I-DAC + R_T	16.51	3.0
Fine Comparator	38.15	7.1
Total	539.88	100
Estimated SNR	57.73 dB	ADC Input 1.6V_{PP}

This switchback operation increases the control logic delay, which degrades the conversion speed, as well as the power efficiency. Fig. 11(b) shows the proposed split current source DAC that eliminates the switchback operation in the two-level referenced DAC. Each binary-weighted current source of the conventional DAC is split in half, with top and bottom cells, similar to the split capacitor DAC in [16]. For MSB conversion, a differential zero is generated by connecting all the bottom cells to the negative output, while all the top cells are connected to the positive output. When the MSB is determined to be one, only the $8I_{UNIT}$ in the bottom cells switches to the positive output for the next reference generation.

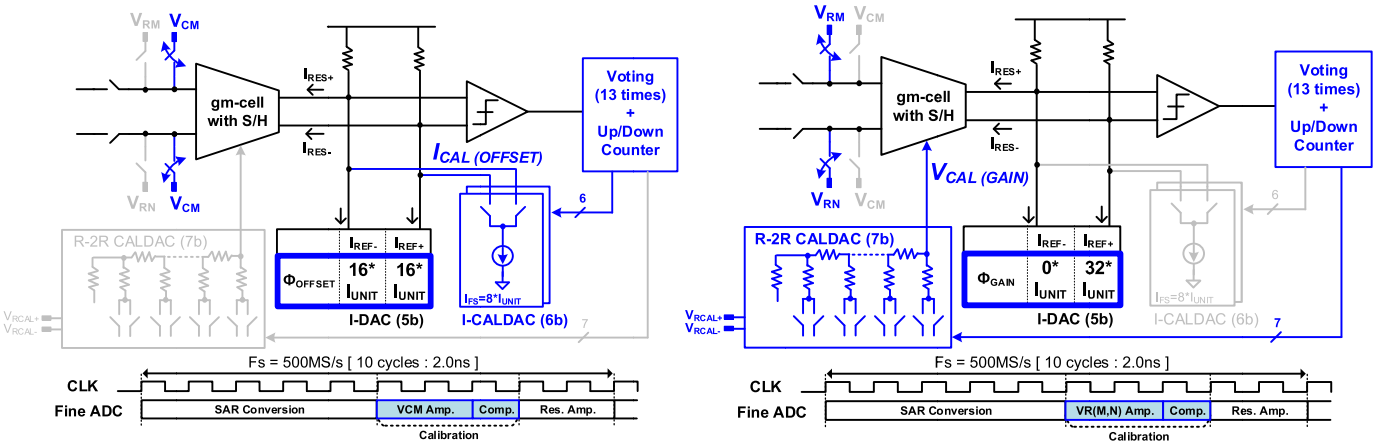
As shown in Fig. 5, owing to the small R_T, all the current sources in the current-steering DAC are designed with sufficient voltage headroom for high linearity even under a 1.0-V supply. The unit current cell of the 400-μA-full-scale 5-bit I-DAC consists of a current source (M_{CS}), a cascode transistor (M_{CAS}), and a switching transistor (M_{SW}, M_{SWB}). The size of the unit current source (M_{CS}) is designed with a sufficient margin to achieve a relative standard deviation of $<2.7\%$, corresponding to about an 8-bit static linearity [18]. With the given voltage swing level at the output of the DAC, the stacked transistors in the saturation region (M_{CAS}, M_{SW}, and M_{SWB}) achieve 100 times larger output resistance than the required value of 160 kΩ [19]. The trans-conductance of the unit current source (M_{CS}) is designed to be as low as possible for the noise performance of the fine ADC. With small switch transistors (M_{SW}, M_{SWB}) and a compact layout of the DAC [20], this work minimizes the parasitic capacitance (C_P) for high-speed residue amplification and high-speed SAR operation.

C. Speed and Noise Analysis

The dominant pole of $1/(R_T \cdot C_P)$ is the common factor that determines the settling time of the gm-cell in the sampling phase and that of the I-DAC for SAR conversion. As shown in Fig. 3, the time of one cycle of the synchronous clock is 200 ps. As the comparator latching and digital logic in the 5-bit fine SAR ADC take about 150 ps in each settling, 50 ps can be assigned for I-DAC settling. Thus, with R_T of 320 Ω, C_P must be smaller than 45 fF. The designed (estimated) parasitic capacitance (C_P) in Fig. 5 is about 30 fF, consisting of 15-fF routing capacitance and 15-fF device parasitic including 9 fF by the 5-bit I-DAC. The resultant small time-constant ($R_T \cdot C_P$) of 9.6 ps also enables fast residue settling of about 250 ps even with the internal parasitic of the gm-cell with S/H, making two clock cycles assigned for residue amplification to be reasonable.

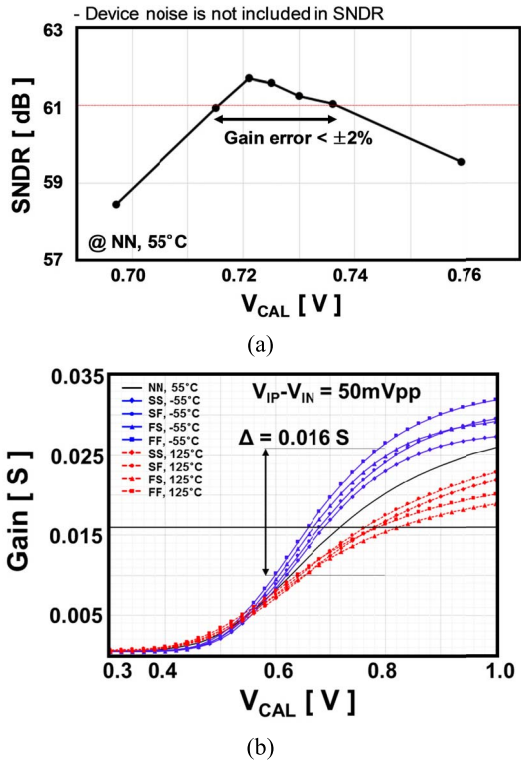
Fig. 12 shows the noise analysis result of the gm-cell with S/H. The noise of the gm-cell is composed of those from the FVF, the sampling network (switch and C_G) in sampling mode (Φ_{IS} = H) and additional noise from the V–I and its output parasitic capacitance (C_P) in holding mode ($\Phi_{IS} = L$). The simulated input-referred noise of the gm-cell with S/H is 180.67 nV^2, mainly caused by the noise of the FVF and the sampling switch. Although a large value of C_G could

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(a) (b)

Fig. 13. Background calibration (a) in offset and (b) in gain.



(a)

(b)

Fig. 14. Simulation results of (a) ADC's signal-to-noise-and-distortion ratio (SNDR) versus V_{CAL} in the gm-cell and (b) gain versus V_{CAL} in the gm-cell.

improve the noise performance of the gm-cell, it degrades the operating speed. The C_G, 15 fF, is implemented with only gate capacitance of the mirror transistor (M5, M6 in Fig. 5) to maximize the speed while the noise of the gm-cell is smaller than the quantization noise in ADC's SNR. Table I shows the simulated noise contribution of the overall ADC. Owing to the noise optimization in the fine ADC, the noise contribution of the fine ADC even with small parasitic capacitance (C_P) is only 10.1% of the total. Hence, the proposed ADC could be designed to achieve an SNR higher than 55 dB with no band-limiting additional capacitors for high speed.

TABLE II
SIMULATED OFFSET VALUE

Current-mode Back-end	Current output offset	Input-referred offset
gm-cell	30.6 μA	1.22 LSB
gm-cell + I-DAC	32.1 μA	1.28 LSB
gm-cell + I-DAC + Comp.	38.4 μA	1.54 LSB

D. Background Calibration

With open-loop residue amplification, as is well known, inter-stage errors should be addressed. In this architecture, the gain mismatch between the full-scale current of the gm-cell and that of the fine current-steering DAC should be calibrated. In this work, a zero input and the positive residue full-scale are taken as two reference points for two-point gain calibration, which inherently corrects the offset error. The proposed two-point gain calibration scheme is conducted in the background with an on-chip calibration block. Both the gain and offset calibration operations share the same clock periods in an alternating manner, as illustrated in the timing diagrams in Fig. 13(a) and (b).

As shown in Fig. 13(a), the offset error is calibrated by applying a differential zero input to the gm-cell after the fine A/D conversion is completed. During this phase, the 5-bit I-DAC in the fine SAR ADC is reset for zero output. The comparator output then indicates the polarity of the combined offset of the gm-cell and I-DAC. By adding a current-mode calibration DAC (I-CALDAC) at the output of the I-DAC, the offset is adjusted by the offset canceling current, I_{CAL}, by referring to the comparator output. Table II shows the Monte-Carlo simulated offset values of the gm-cell and fine ADC. The 1 sigma value (σ) of the total current offset of the current-mode back-end circuit (before calibration) is 38.4 μA, dominated by the effects of the gm-cell and the comparator. Based on the simulated input-referred offset of a 1.54 LSB, a 6-bit I-CALDAC is designed with a differential full-scale current of 200 μA (coverage of 5.2 sigmas), which can adjust the input-referred offset with a step size of 0.125 LSB.

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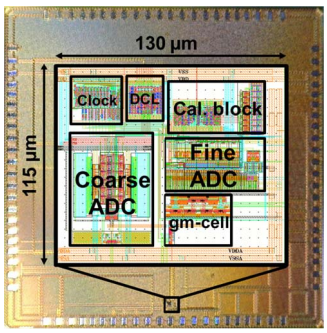


Fig. 15. Die photograph and layout.

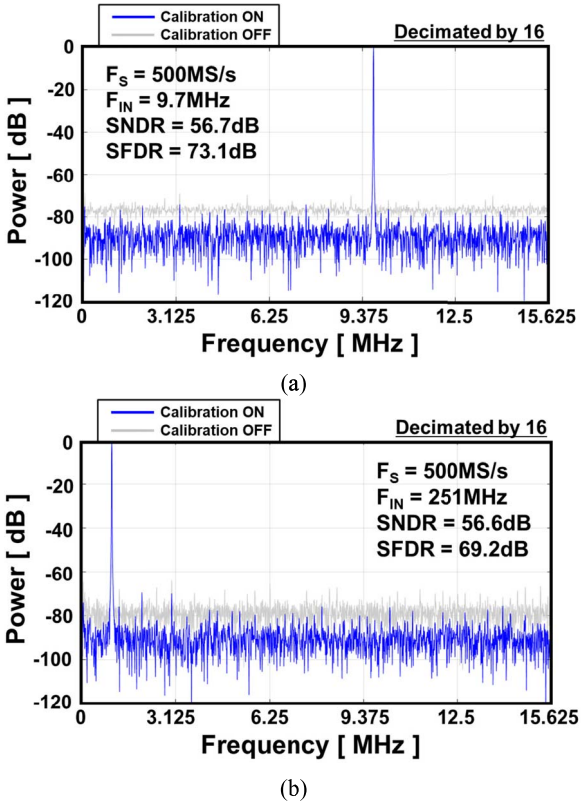


Fig. 16. Measured FFT spectrum at 500 MS/s with (a) 9.7- and (b) 251-MHz input.

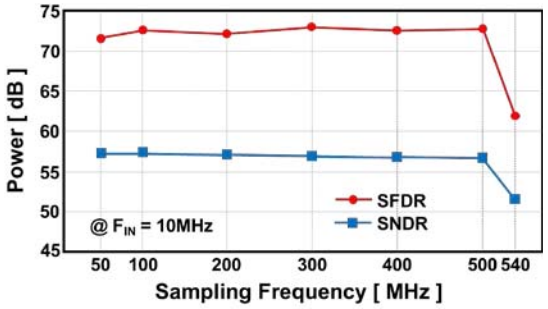


Fig. 17. Measured SNDR and spurious-free dynamic range (SFDR) versus sampling frequency.

For gain error calibration, as shown in Fig. 13(b), the gm-cell is connected to the reference voltages that represent the full-scale residue, V_{RM} and V_{RN}, and the 5-bit I-DAC is set

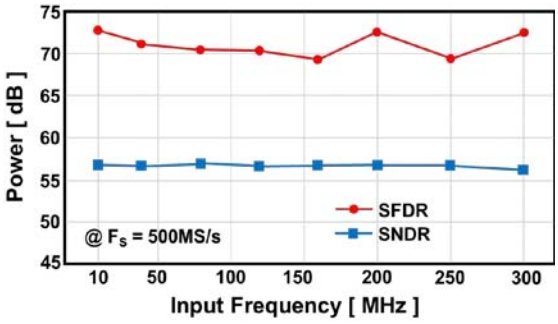


Fig. 18. Measured SNDR and SFDR versus input frequency.

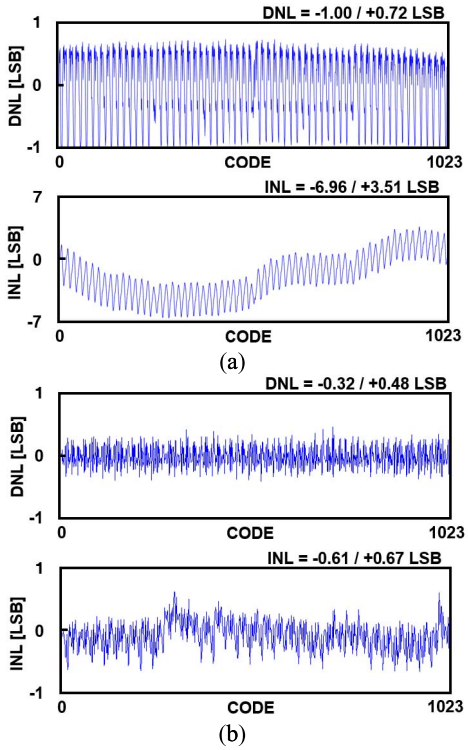


Fig. 19. Measured differential non-linearity (DNL) and integral non-linearity (INL) (a) before calibration and (b) after calibration.

to its full-scale value ($32*I_{UNIT}$). An additional calibration DAC (7-bit R-2R CALDAC) adjusts the degeneration resistor value (MR1 in Fig. 5) of the gm-cell with V_{CAL} so that the output current from the gm-cell converges to the full-scale current of the I-DAC. For an SNDR (without device noise) over 61 dB, as simulated in Fig. 14(a), the allowed gain error range is less than $\pm 2\%$ of the full-scale current of the I-DAC. Fig. 14(b) shows the voltage-to-current conversion gain change according to the V_{CAL} in the various process corners and temperature conditions. The worst-case simulated value of the gain variation is about 0.016 S at the edge of V_{CAL} of 0.7 V. Hence, the 7-bit R-2R CALDAC has a control range of about 200 mV and a step size below 1% for gain control. The two references, V_{RM} and V_{RN}, are generated utilizing a resistive divider from an additional ADC reference, considering a residue gain drop of about 20% due to parasitic

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TABLE III
PERFORMANCE SUMMARY AND COMPARISON

	VLSI16 Y.-C. Lien [8]	VLSI13 B. Verbruggen [6]	TVLSI17 Y. Zhu [7]	JSSC17 H. Huang [9]	ISSCC18 L. Kull [10]	This Work
Technology	28 nm	28 nm	65 nm	65 nm	14 nm	28 nm
Resolution [bit]	12	11	10	12	10	10
Residue amp. Structure	Gain-boosting telescopic amplifier	Dynamic amplifier	Gain-boosting telescopic amplifier	Dynamic amplifier	CS amplifier	gm-cell
F_S [MS/s]	800	410	500	330	1500	500
# of TI channels	4	2	2	-	-	-
F_S per channel [MS/s]	200	205	250	330	1500	500
Supply Voltage [V]	1.0	1.0	1.2	1.2	0.95	1.0
SNDR @Nyquist [dB]	60.8	55	55.9	63.5	50.1	56.6
Power [mW]	14.6	2.1	8.2	6.23	6.92	6.0
FoM_{Walden} @Nyq. [fJ/conversion-step]	20.4	11.4	32.2	15.4	17.7	21.7
$FoM_{Schreier}$ @Nyq. [dB]	165.2	164.9	160.7	167.8	160.5	162.8
Area [mm^2]	0.093	0.11	0.046	0.08	0.0016	0.015

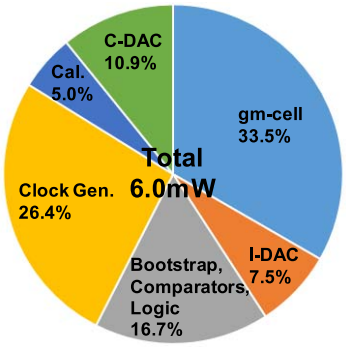


Fig. 20. Power consumption breakdown.

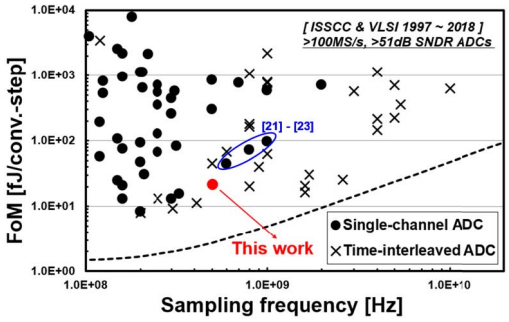


Fig. 21. FoM comparison with published ADCs [24].

capacitance at the top-plate node of the coarse ADC. Thirteen times of voting are conducted with the comparator decisions for a single decision of error polarity for improved calibration accuracy. Note that the required input-reset of the gm-cell after calibration is actually conducted during the "Comp" period in Fig. 13(a) and (b) as the FVF is disconnected from the V–I in hold mode.

IV. MEASUREMENT RESULTS

A prototype pipelined-SAR ADC with the current-mode residue processing was implemented in a 28-nm CMOS process. Fig. 15 shows the chip photograph with an active area of 0.015 mm^2. Additional hardware involved in the calibration for inter-stage mismatch errors accounts for approximately 10% of the total area of the ADC, as shown in the layout. All the measured results are achieved at a 500-MS/s conversion rate and a 1.0-V power supply. Fig. 16(a) and (b) show the measured fast Fourier transform (FFT) spectrums decimated by a factor of 16 with and without background calibration. When calibrations for both offset and gain error are enabled, at a 9.7-MHz input, the SNDR and SFDR of the ADC are 56.7 and 73.1 dB, respectively. With an input at 251 MHz, the measured SNDR and SFDR are 56.6 and 69.2 dB, respectively. Fig. 17 shows the ADC's dynamic performance for various sampling rates with a 10-MHz input, presenting an SFDR higher than 72 dB up to a 500-MS/s conversion rate. Fig. 18 shows the dynamic performance for various input frequencies at a 500-MS/s sampling rate. It shows quite a constant performance, maintaining an SNDR above 56 dB up to the input frequency of 300 MHz. Fig. 19 shows the linearity performances before and after calibration. Before calibration, the DNL shows many missing codes. After calibration is enabled, the DNL and INL are −0.32/+0.48 and −0.61/+0.67 LSB, respectively. As the power breakdown presented in Fig. 20, the total power consumption is 6 mW, including 2 and 1.6 mW by the gm-cell and the clock buffer, respectively.

Table III summarizes the performance of the prototype ADC and compares it with previously reported high-speed pipelined SAR ADCs. This prototype ADC achieves the highest single-channel conversion-rate among ADCs with various residue amplification techniques, except for the case of the FinFET process [10]. These results show that the proposed ADC with a current-mode residue processing scheme is a promising pipelined-SAR architecture for high-speed

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applications. Fig. 21 compares the figure of merit (FoM) of this work with previously reported ADCs. Note that this work achieves low power consumption with a faster residue processing period (<400 ps) relative to other high-speed pipeline ADCs [21]–[23], showing a remarkable FoM of 21.7-fJ/conversion-step among the published single-channel ADCs, which are indicated with black dots.

V. Conclusion

This paper introduces a current-mode open-loop residue processing technique with a SAR-based pipelined architecture. The residue processing in the current domain enhances the operation speed in the pipelined SAR ADC, owing to the low-impedance signaling and the high-speed split-current source DAC switching. An area-efficient background calibration function is embedded on a chip to resolve the inter-stage mismatch issues in the open-loop voltage-to-current conversion. The single-channel 10-bit 500-MS/s prototype ADC achieving an FoM of 21.7-fJ/conversion-step successfully verified the design concept.

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Received XX Month XXXX; revised X Month XXXX; accepted XX Month XXXX. Date of publication XX Month XXXX; date of current version XX Month XXXX.

Digital Object Identifier 10.1109/ICAS.2020.2976889

A 64-Gb/s 0.33-pJ/bit PAM4 Receiver Analog Front-End with a Single-Stage Triple-Peaking CTLE Achieving 22.5- dB Boost in 40-nm CMOS Process

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This woik was supported by the National Natural Science Foundation of China under grant62222409 and 62174153.

ABSTRACT This work presents a PAM4 receiver analog frontend (AFE) operating up to 64 Gb/s. The electronic integrated circuit (EIC) is fabricated in 40-nm CMOS technology. This AFE is composed of a single-stage Continuous-Time Linear Equalizer (CTLE), a Variable Gain Amplifier (VGA), an input impedance matching network, a buffer stage, and an output buffer. The single-stage triple-peaking CTLE proposed employs current reuse technique and a multi-feedback structure, enabling the adjustment of peaking in the low, mid, and high-frequency bands. Thus, only one-stage CTLE is sufficient to achieve an over-20-dB boost at Nyquist frequency to save power. The VGA adopts an enhanced structure based on the Gilbert cell, where the gain is manipulated by controlling the gate voltage of MOS transistors. The CTLE undergoes variations in its DC gain during the adjustment process to equalize channel losses. The role of the VGA is to stable the DC gain changes induced by the adjustment of the CTLE. The output buffer adopts two stages, aiming to ensure that the gain does not attenuate excessively while maintaining output impedance matching. The AFE consumes 21.1mW with a supply voltage of 1.5/1V. It can provide a maximum boost of 22.5 dB, and the data rate reaches up to 64 Gb/s. Additionally, it features peaking adjustment capabilities in the low, mid, and high-frequency bands. Finally, the measurement demonstrates its ability to effectively equalize a channel with a 12-dB loss at the Nyquist frequency of 16 GHz.

INDEX TERMS Analog front-end (AFE), continuous-time linear equalizer (CTLE), VGA, Triple Peaking, PAM4, low-power

I. INTRODUCTION

In recent years, with various breakthrough applications, such as cloud computing, the Internet of Things (IoT), artificial intelligence, etc., the data throughput of data centers has been steadily increasing year by year[1,2]. Pulse-amplitude modulation (PAM-4) signaling continues to be the preferred modulation scheme, primarily due to its superior spectral efficiency compared to non-return-to-zero (NRZ) modulation and its more straightforward implementation compared to more complex schemes[3]. Therefore, enhancing the energy efficiency of the widely utilized PAM4 serial link transceiver, crucial for data transmission within data centers, holds significant importance in addressing the power consumption challenges posed by the expanding scale of data centers[4,5]. In PAM4 serial link transceivers, a substantial portion of power consumption is attributed to the analog front-end (AFE) circuit. This is due to its requirement to compensate for channel loss through the continuous-time linear equalizer (CTLE) and amplify the swing of the CTLE output to several hundreds of mV. Therefore, the low-power AFE circuit holds significant importance in the design of low-power PAM4 serial link transceivers.

The primary function of the AFE is to equalize the losses of the channel. We understand that channels introduce high-frequency losses due to phenomena like skin effect, which are non-ideal characteristics. However, for channels in certain mediums, significant attenuation can also occur at low and mid frequencies. This necessitates the CTLE in the AFE circuit to have adjustable compensation capabilities across low, mid, and high-frequency bands for better equalization of channel losses. In previous works, conventional CTLE only achieves high-frequency peaking by reducing DC gain. In order to increase the equalization capability of CTLE, low-frequency feedback technology is used to weaken the DC gain before fixing low-frequency, and then high-frequency feedforward technology is used to achieve high-frequency boost adjustment. Combining technologies above can achieve peaking adjustment at low, medium and high frequency. However, the topology need for multi-stage CTLE cascading, leading to higher power consumption[6,7].

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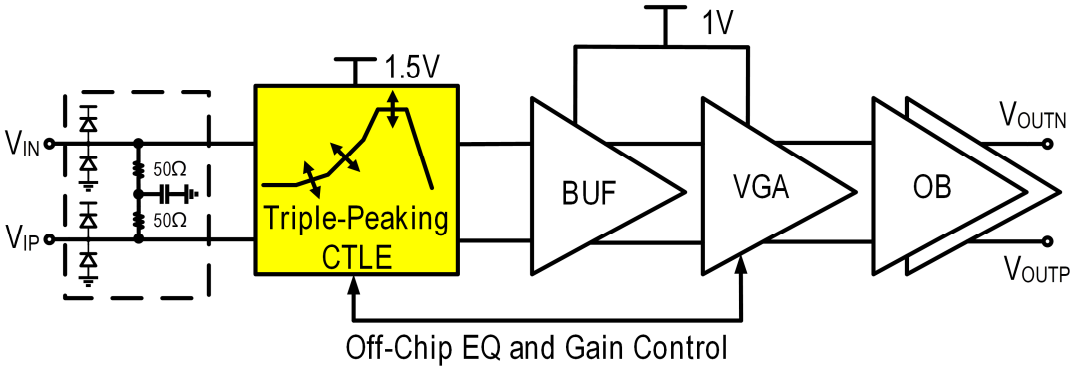


FIGURE 1. **The Proposed Receiver AFE Architecture.**

To surmount this issue, in this article, we present a 64-Gb/s PAM4 receiver AFE designed in a 40-nm CMOS process, achieving extremely low energy efficiency of 0.33pJ/bit[8]. To reduce AFE power consumption, which was predominantly associated with multi-stage CTLE in prior research, we present a groundbreaking solution—a single-stage triple-peaking CTLE (STP-CTLE). This inventive design integrates a current reuse technique, enabling the manipulation of three peaks (low, middle, and high-frequency peaking) with only a single stage. A buffer is connected after the CTLE to stabilize the output common-mode voltage, and simultaneously, it can drive larger input capacitance in the subsequent VGA. The VGA is an improved version based on the Gillbert cell structure. In contrast to the traditional Gilbert cell, which adjusts gain by varying the current source magnitude, the proposed VGA modifies gain by altering the gate voltage of the MOS transistors in series with the load resistance. This structure exhibits superior linearity, making it particularly advantageous for PAM4 signals that demand high linearity[9]. The output buffer employs a two-stage buffer structure, enabling impedance matching without compromising the overall gain performance of the AFE.

II. Receiver AFE Architecture

Fig. 1 shows the overall block diagram of the AFE, consisted of an input impedance matching network, a STP-CTLE for compensating channel losses, a VGA for gain control, a buffer isolating between CTLE and VGA, and an output buffer for testing purposes.

The STP-CTLE can achieve a maximum channel compensation of 22.5dB. It has adjustable compensation capabilities across low, mid, and high-frequency bands. Additionally, due to the adoption of current reuse technology, its power consumption is only 12.6mW. The gain adjustment range of the VGA is 5.8dB. As the CTLE adjusts its parameters to compensate for channel variations, there are changes in the DC gain of the CTLE. To counteract this, we appropriately adjust the parameters of the VGA to ensure that the overall gain, considering both the VGA and CTLE, remains constant. The addition of a buffer between VGA and CTLE serves two purposes. Firstly, to accommodate the need for VGA to provide a larger gain, its input transistors have larger dimensions, resulting in increased parasitic capacitance at the input. Introducing a buffer enhances the drive capability of the preceding stage to VGA, ensuring sufficient bandwidth. Secondly, when CTLE adjusts the magnitude of high-frequency peaking, it alters the DC output operating point. The inclusion of a buffer is intended to stabilize the DC operating point. Additionally, the output buffer (OB) is used for testing, which employs a highfrequency peaking to compensate for the loss of testing cable.

III. Proposed Single-Stage Triple-Peaking CTLE

In previous Analog Front-End (AFE) designs, achieving three peaks necessitated the use of at least two cascaded CTLEs, as depicted in Fig 2. The first stage could employ a high-frequency feedforward CTLE (HFF-CTLE), integrating a degenerated RC of the source and a high-frequency feedforward topology to achieve mid and high-frequency peaking. Meanwhile, the second stage, utilizing a low-frequency equalization (LFEQ) [10], could incorporate a feedback loop with a low-pass filter for low-frequency peaking. As illustrated in Figure 2, such a two-stage CTLE configuration faced challenges related to significant power consumption, primarily due to the intricate circuit topology. In the HFF-CTLE, the high-frequency feedforward module consumes a current I_H, and the main input transistor current consumption is $2I_M$. To ensure an adequate adjustment range for high-frequency peaking, the tail current source I_H should not be significantly smaller than $2I_M$. In the LFEQ module, as I_L is in the feedback path, a smaller current source is needed. The I_{BUF} influences the output common-mode voltage and swing and should be comparable in magnitude to $2I_M$. Then, the power consumption of the traditional implementation with two stages of CTLE for achieving triple peaking is: $(I_H+4I_M+I_L+I_{BUF})\times 1V$, which is substantial for low-power AFE designs.

A. Circuit Description

To address this concern, we suggest consolidating three peaks into a single-stage CTLE, aiming to streamline the

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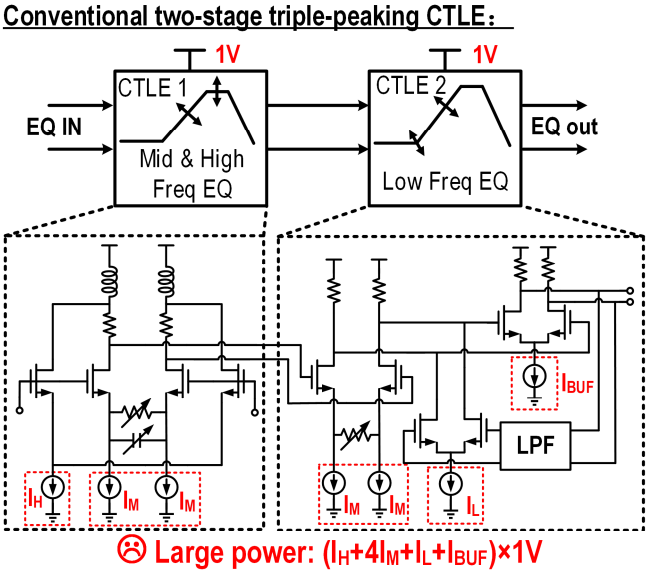


FIGURE 2. **Conventional two-stage triple-peaking CTLE.**

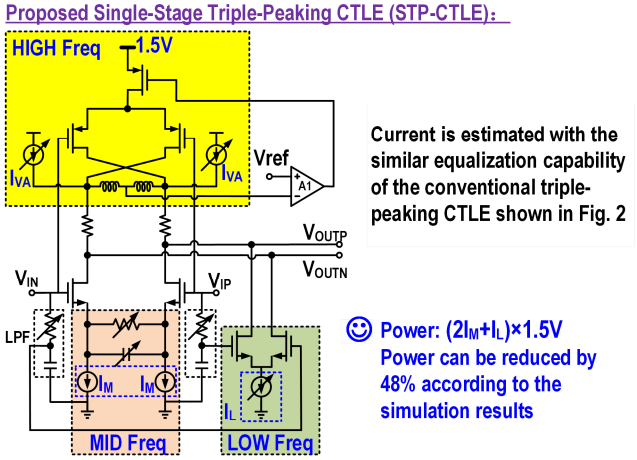


FIGURE 3. **Proposed STP-CTLE.**

circuit configuration, as demonstrated by our proposed STP-CTLE illustrated in Fig. 3. The high-frequency boosting component (HFB) of our STP-CTLE primarily incorporates a PMOS feedforward differential pair, effectively leveraging the entire current I_{TOT} ($I_{TOT}=I_M\times2+I_L$, see Fig. 3) of the STP-CTLE, thus leading to a significant power consumption reduction. High-frequency feedforward transistors act on the high-frequency gain through inductance without affecting the low-frequency gain[11]. As indicated in Fig. 3, since I_{TOT} also equals to (I_P+2I_{VAR}), the current of the PMOS differential pair I_P can be adjusted by tuning the current I_{VA} of two adjustable current sources. So, the transconductance of the PMOS differential pair in HFB part can be adjusted, thus the magnitude of the highfrequency peaking can be controlled. An operational amplifier A1 is used to form a common-mode feedback loop. Then, the point between the two inductors can be considered as a virtual ground. This allows the output of HFB to be directly coupled to the overall output of the STP-CTLE.

The mid-frequency peaking part (MFB) mainly consists of an RC source-degenerated differential pair. It adjusts the source-degenerated resistor and capacitor to tuning the position and magnitude of mid-frequency peaking[12]. In this setup, we employ a series resistor array and a parallel capacitor array to achieve the adjustment of resistance and capacitance values[13]. Additionally, increasing the source degeneration resistance in MFB results in a reduction of its transconductance (gm), which can enhance the linearity of the CTLE[14].

The foundation of the low-frequency boosting part (LFB) lies on a feedback loop, comprising a differential pair and a series of cascaded low-pass filters (LPF) integrated with a variable resistor R_{VAR}[15]. The tail current source of the differential pair is designed as an adjustable current source. Altering the size of the current source can modify the transconductance (gm) of the differential pair, thereby adjusting the strength of the low-frequency feedforward. By increasing the tail current I_L of LFB, the DC gain of the whole CTLE gets decreased while the high-frequency gain is not affected, thus reducing the low-frequency boosting. Moreover, the signal passes through a pair of RC low-pass filters before entering the gate of the differential pair. Consequently, low-frequency feedforward theoretically only acts on signals below the cutoff frequency of this low-pass filter (LPF). Since the resistance of the LPF is designed as an adjustable resistor array, altering the resistance values will change the cutoff frequency of the LPF, consequently influencing the location of low-frequency peaking.

Since the number of the stacked MOSFET is increased, the supply voltage for our STP-CTLE has been raised from 1 V to 1.5 V to ensure ample voltage headroom[13]. However, The power consumption of our proposed STP-CTLE is ($2I_M+I_L$) ×1.5V, which represents approximately a 48% reduction compared to the power consumption of the traditional multi-stage CTLE structure. This significant reduction is primarily attributed to the substantial decrease in current consumption within the STP-CTLE, which consequenced by the adoption of current-reuse techniques effectively saving the current overhead associated with the high-frequency feedback module. It's important to note that the power consumption comparison between the two is based on simulation results, assuming comparable equalization capabilities and bandwidth for both structures.

B. Zero and pole analysis

Fig. 4 depicts the small-signal equivalent circuit of the LFB and MFB for analyzing the locations of zeros and poles within its low-frequency and mid-frequency domain. The cutoff angular frequency of the Low-Pass Filter (LPF) in the figure is

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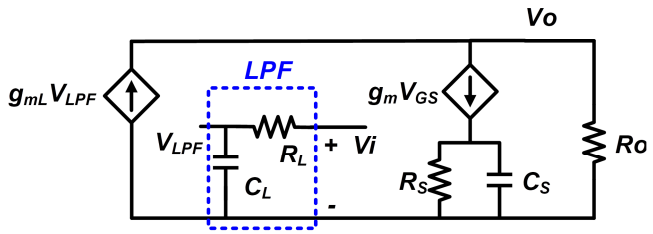


FIGURE 4. **The small-signal equivalent circuit of LFB & MFB.**

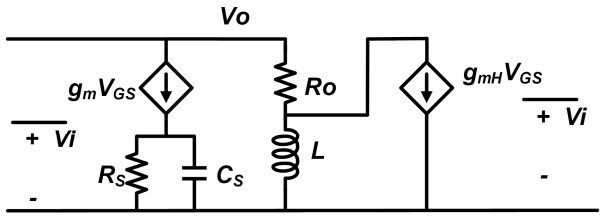


FIGURE 5.**The small-signal equivalent circuit of HFB & MFB.**

$$\omega_c = \frac{1}{R_L C_L} \tag{1}$$

The input V_{LPF} of LFB is obtained by passing V_i through the LPF. Therefore, we can straightforwardly deduce the pole generated by LFB to be

$$\omega_{p1} = \omega_c = \frac{1}{R_L C_L} \tag{2}$$

Then, by calculating the frequencies at which the sum of LFB and MFB outputs is zero, we can determine the zeros generated by LFB to be

$$\omega_{z1} = \frac{g_{mL}}{C_S(g_m - g_{mL})} - \frac{1}{R_S C_S} \tag{3}$$

The zeros and poles introduced by the source degeneration RC in MFB occur in the mid-frequency range, and they are given by:

$$\omega_{z2} = \frac{1}{R_S C_S},\ \omega_{p2} = \frac{1+g_m R_S}{R_S C_S} \tag{4}$$

Fig. 5 represents the small-signal equivalent circuit of HFB. Due to the introduction of inductance, the analysis of zeros and poles involves solving multiple equations. Assuming only HFB and MFB are presented (which does not significantly impact the zero-pole analysis since, despite the shared current, the outputs of HFB and LFB are superimposed on MFB, and there is no direct interaction between HFB and LFB), we obtain the transfer function for this scenario as follows:

$$H(s) = -\left[\frac{g_m(R_O+sL)}{1+g_m(R_S//\frac{1}{C_S})} + g_{mH}sL\right] \times \frac{1}{LC_L s^2 + R_O C_L s + 1} \tag{5}$$

It's worth noting that, as we analyze the HFB and MFB modules, there is a pair of zeros and poles here that coincide with the mid-frequency zeros and poles in LFB & MFB, denoted as ω_{z2} and ω_{p2}. Due to the inductance L and the load resistance R_O, forming an RLC model with the load capacitance C_L, two high-frequency poles will be generated. Additionally, gm_H introduces a high-frequency zero.

In equations (1)~(5), R_S and Cs represent the source degeneration resistance and capacitance of the equivalent half-circuit, respectively; gm, gm_L, and gm_H are the transconductance of the differential pairs of MFB, LFB and HFB, respectively; L is the inductance value, and C_L is the load capacitance of the CTLE.

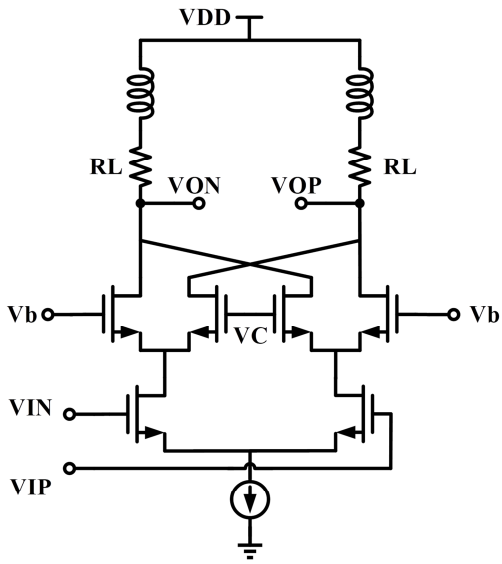


FIGURE 6. **The Circuit diagram of the Variable Gain Amplifier.**

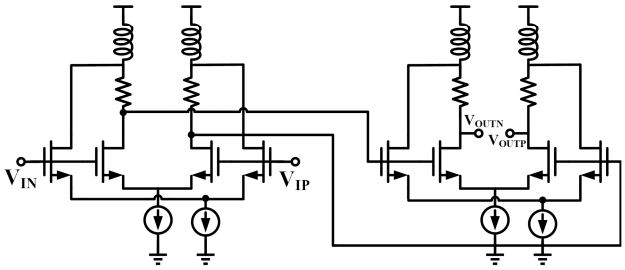


FIGURE 7. **The Circuit diagram of the Output Buffer.**

IV. Other Blocks

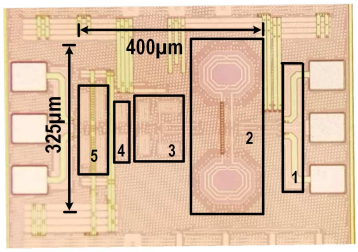
A. Variable Gain Amplifier

The Variable Gain Amplifier (VGA) adopted is an improved version based on the Gilbert cell as shown in Fig. 6. The traditional VGA structures include the Gilbert cell structure, where gain is controlled by adjusting the tail current source size, but it is challenging to achieve good linearity. Another approach is the source-degenerated differential pair, however, it suffers from significant bandwidth and group-delay variation across the gain settings. The VGA based on Transimpedance Amplifier (TIA) adjusts gain by changing the feedback resistor, but it has the drawbacks of structural complexity and high power consumption.

The VGA circuit in Figure 6 controls the conduction resistance of a pair of cross-coupled transistors by the voltage VC. Consequently, the magnitude of the alternating current flowing through the load resistor can be controlled.

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This article has been accepted for publication in Integrated Circuits and Systems. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.23919/ICS.2024.3456043



Power (mW)	
STP CTLE	12.633
BUF	4.0
VGA	4.482
Total	21.1m W

FIGURE 8. Chip photograph with power consumption breakdown.

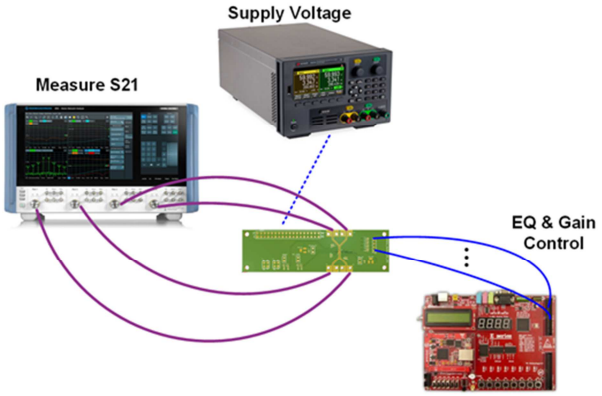


FIGURE 9. The platform for Chip testing.

When the voltage VC increases, the conduction resistance of the cross-coupled transistors decreases, resulting in a smaller alternating current through the load resistor RL, and hence, a smaller gain. The voltage VC is regulated externally using a Low Dropout Regulator (LDO).

B. Output Buffer

The output buffer is divided into two stages. The purpose of the first-stage pre-buffer is to ensure that the overall gain of the output buffer (OB) is not too small. This is necessary because the second-stage buffer serves to impedance match, having a smaller load resistor, which would result in a lower gain.

V. EXPERIMENTAL RESULTS

The proposed PAM4 AFE is fabricated in a 40-nm CMOS technology, occupying a core area of 0.136 mm2, as shown in Fig. 8. The measured power of the AFE except OB for testing is 21.1mW, corresponding to an energy efficiency of 0.33 pJ/bit. Fig. 9 illustrates the testing setup for the AFE chip, with S-parameters measured using a vector network analyzer. The internal parameters of the chip are controlled by an FPGA. The S21 parameter of the AFE, as shown in Figure 10, demonstrates that triple-peaking is achieved in a single stage CTLE with a maximum boost of 22.5-dB at 16-GHz Nyquist frequency.

As mentioned earlier, the magnitude of low-frequency peaking in the proposed AFE is controlled by the current source IL and the resistor RVAR in the LPF, while the

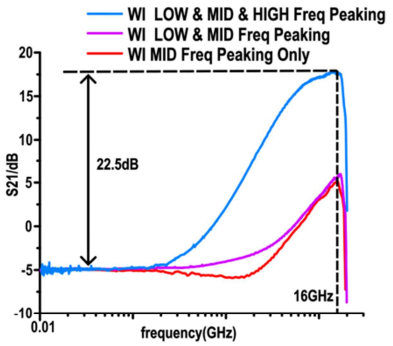


FIGURE 10. Measured S21 of the AFE.

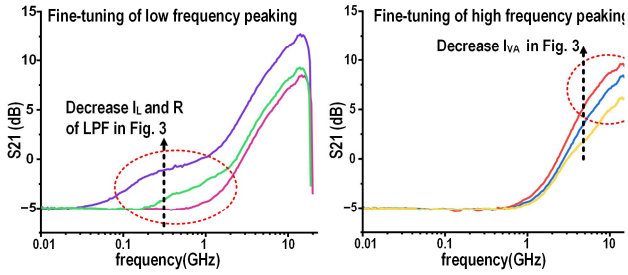


FIGURE 11. Measured fine tuning of low frequency peaking and high frequency peaking.

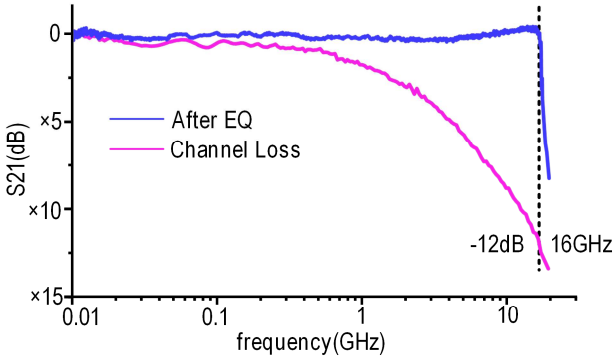


FIGURE 12. Measured AC response before and after channel loss compensation.

magnitude of high-frequency peaking is controlled by the current source IVA. The test results in Figure 11 illustrate the adjustment effects on the low-frequency and high-frequency peaking of the AFE. On the left side of Figure 11, the effects of adjusting IL and RVAR on low-frequency peaking are shown, while on the right side, the effects of adjusting the current source IVA on high-frequency peaking are depicted.

Next, we introduced a channel PCB between the AFE chip input and the vector network analyzer. By adjusting the parameters of the AFE, we obtained the S21 parameter of the AFE after channel equalization and compared it with the S21 of the channel PCB. As shown in Figure 12, the upper curve represents the S21 after equalization, and the lower curve represents the S21 of the channel. The

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This article has been accepted for publication in Integrated Circuits and Systems. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.23919/ICS.2024.3456043

TABLE I
PERFORMANCE COMPARISON

	This work	JSSC'18 [6]	VLSI' 18 [7]	TCAS-II'21 [14]	JSSC' 22 [16]	TCAS-I'22 [17]
Modulation	PAM4	PAM4	PAM4	PAM4	NRZ	PAM4
CMOS (nm)	40	28	16	40	28	65
Data rate(Gb/s)	64	64	56	56	56	56
Supply(V)	1.5/1	1	0.9/1.2/1.8	1.1/1.2	NA	1.1/1.2
Power (mW)	21.1	27$	34	30	43.6	28
Power Efficiency (pJ/bit)	0.33	0.42$	0.61	0.54	0.77	0.52
Equalization	22.5dB	16	17	9	16	14.37
Core Area (mm^2)	0.136	0.32	0.021	0.32	0.102	0.112

$CTLE Only

proposed AFE demonstrates effective channel equalization, achieving a compensation for a 12-dB channel loss at 16 GHz.

The measured performance of the RX AFE module is summarized in Table I and compared against recently published RX AFE for >56 Gb/s in CMOS technology nodes. At 64Gb/s data rate, the presented AFE display a minimum power efficiency of 0.33 pJ/bit. With the Equalization capability beyond 20dB, tht CTLE prove the best energy efficiency and triple-peaking capability.

VI. Conclusion

A PAM4 Receiver Analog Front-End operating up to 64 Gb/s has been presented. The AFE has been confirmed to possess the capability of triple peaking, with the maximum peaking reaching 22.5 dB. The power supply voltages are 1.5V and 1V, where 1.5V is the power supply voltage for STP-CTLE, and 1V is the power supply voltage for the other modules. Thanks to the proposed STP-CTLE, our AFE work achieves better energy efficiency and higher peaking boost at Nyquist frequency.

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